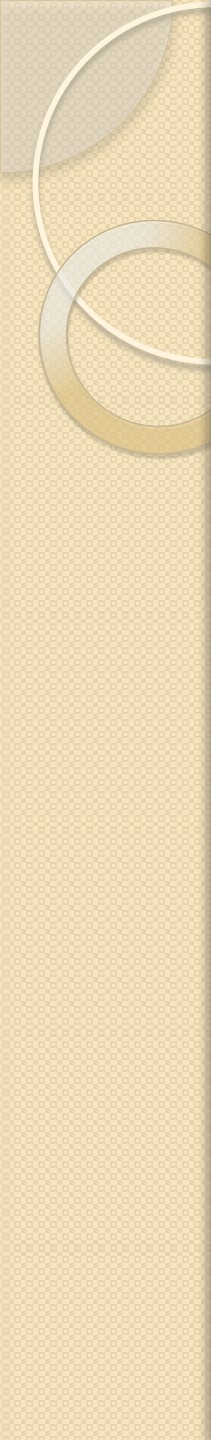
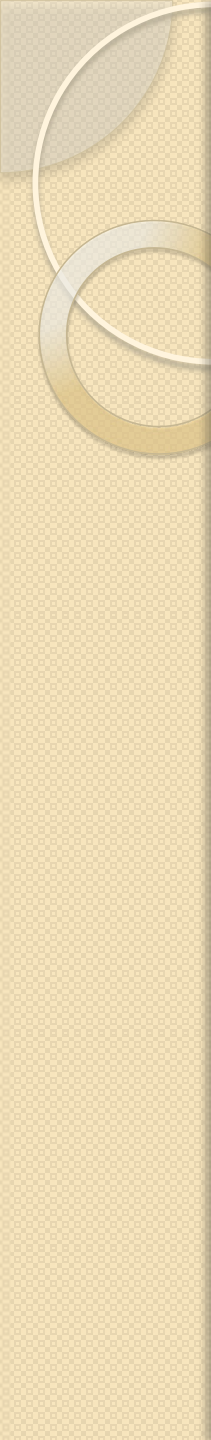


Heart Disease

Overview of Heart Disease

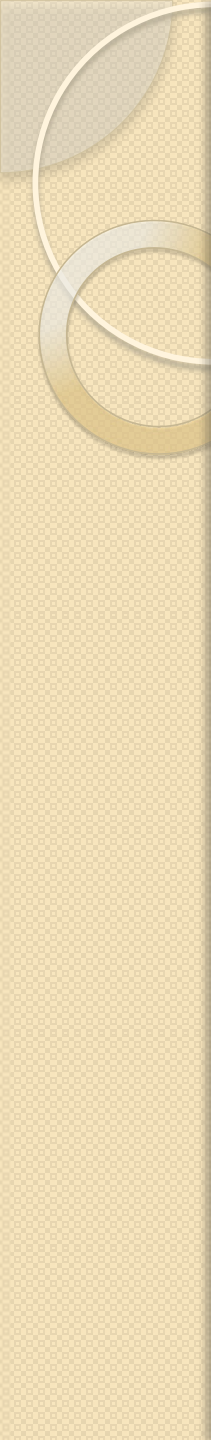
- The heart beats more than 40 million times per year and pumping over **7500 liters** of blood a day
- Cardiovascular disease is the leading cause of mortality worldwide.
- Cardiac dysfunction can be associated with devastating physiologic consequences.
- It can be prevented by lifestyle changes such as quitting smoking, exercising, and following a healthy diet, and correcting its risk factors (such as diabetes, dyslipidemia, and obesity) in their early stages.

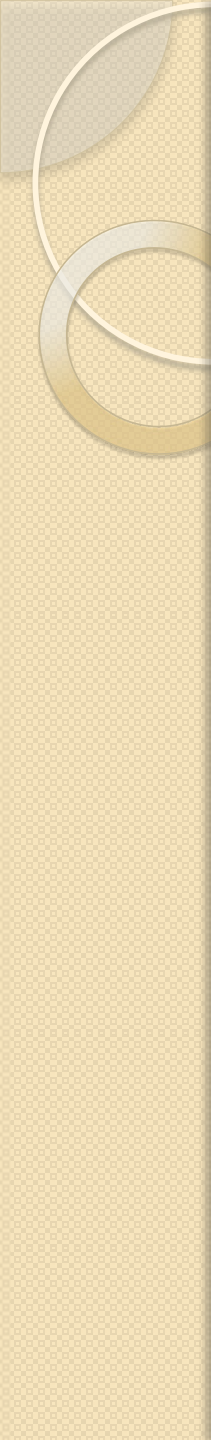
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- IHD can range from asymptomatic coronary heart disease through to stable/unstable angina and myocardial infarction, with several consequences, such as chronic heart failure, arrhythmias, and even death.
 - Valvular heart diseases are also common in practice, taking the forms of **stenosis, insufficiency, or a combination of the two**

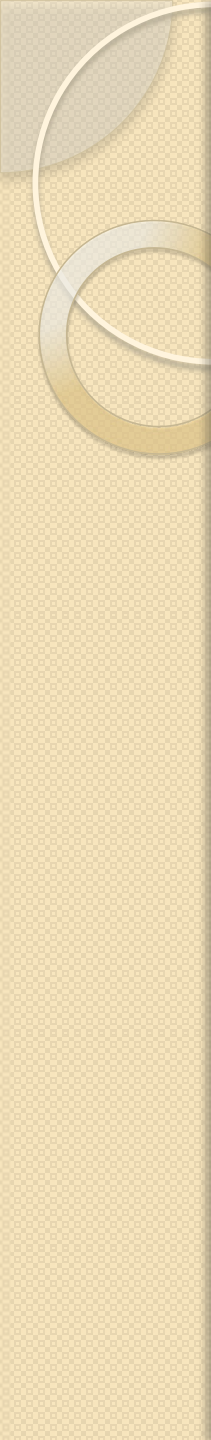
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- The six principal mechanisms of heart diseases:
 - 1) Failure of the pump. In most cases cardiac muscle contracts weakly “systolic dysfunction.” and in some cases, the muscle cannot relax sufficiently (diastolic dysfunction)
 - 2) Obstruction to flow
 - 3) Regurgitant flow
 - 4) Shunted flow
 - 5) Disorders of cardiac conduction. Uncoordinated cardiac impulses or blocked conduction pathways
 - 6) Rupture of the heart or major vessel.

HEART FAILURE

- Often referred to as **congestive heart failure (CHF)**, is the common end point for many forms of cardiac disease and typically is a **progressive condition with a poor prognosis**.
- CHF occurs when the heart cannot generate sufficient output to meet the metabolic demands of the tissues, or can only do so at higher-than-normal filling pressures.
- In a minority of cases, heart failure is a consequence of greatly increased tissue demands, as in hyperthyroidism, or decreased oxygen carrying capacity, as in anemia (**high output failure**).

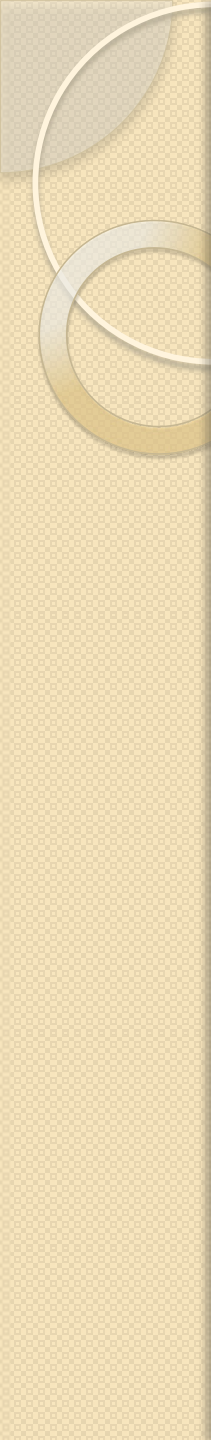
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- In most cases CHF develops gradually and insidiously owing to the cumulative effects of **chronic work overload or progressive loss of Myocardium**
 - Sometimes it is abrupt, as in the setting of a large myocardial infarct or acute valve dysfunction.
 - Heart failure may result from **systolic or diastolic dysfunction**.
 - Approximately one half of CHF cases are attributable to diastolic dysfunction

- 
- Inadequate cardiac output— called **forward failure**—is almost always accompanied by increased congestion of the venous circulation—that is, **backward failure**
 - Congestive heart failure is a multisystem derangement that occurs when the heart is no longer able to eject the blood delivered to it by the venous system

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- The cardiovascular system attempts to compensate for reduced myocardial contractility or increased hemodynamic burden through several homeostatic mechanisms:

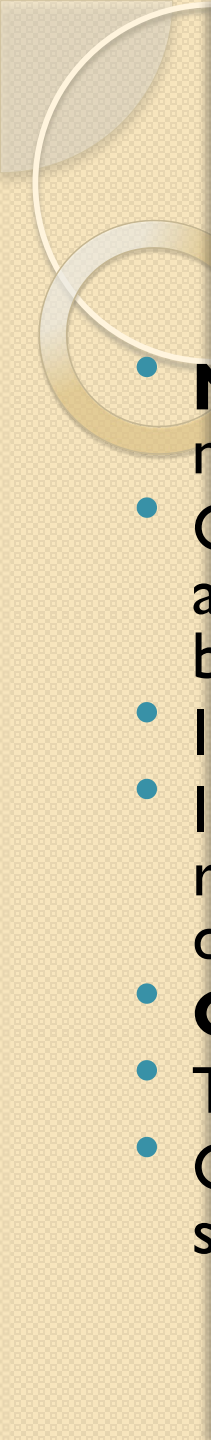
The Frank-Starling mechanism.

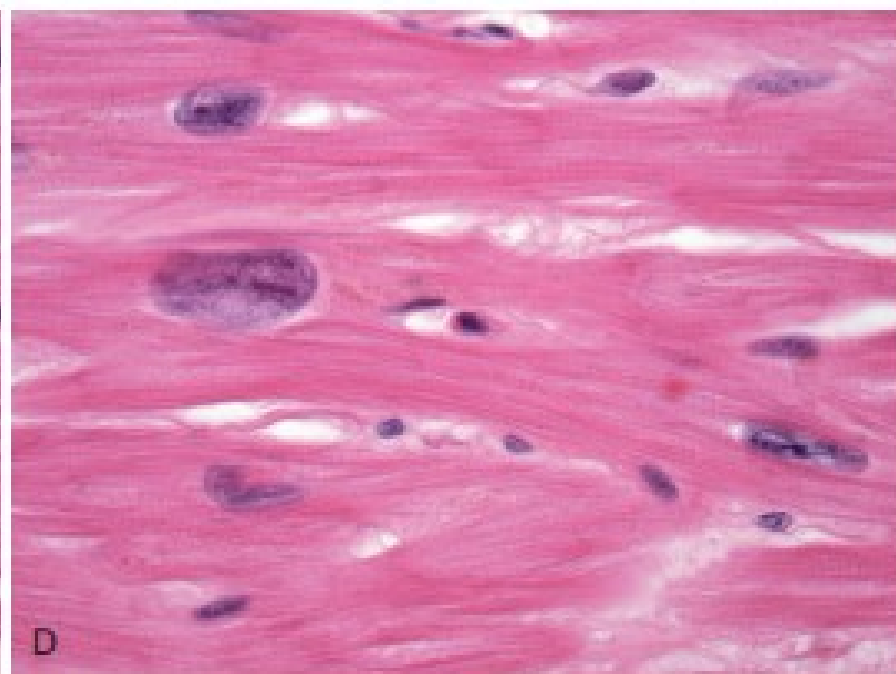
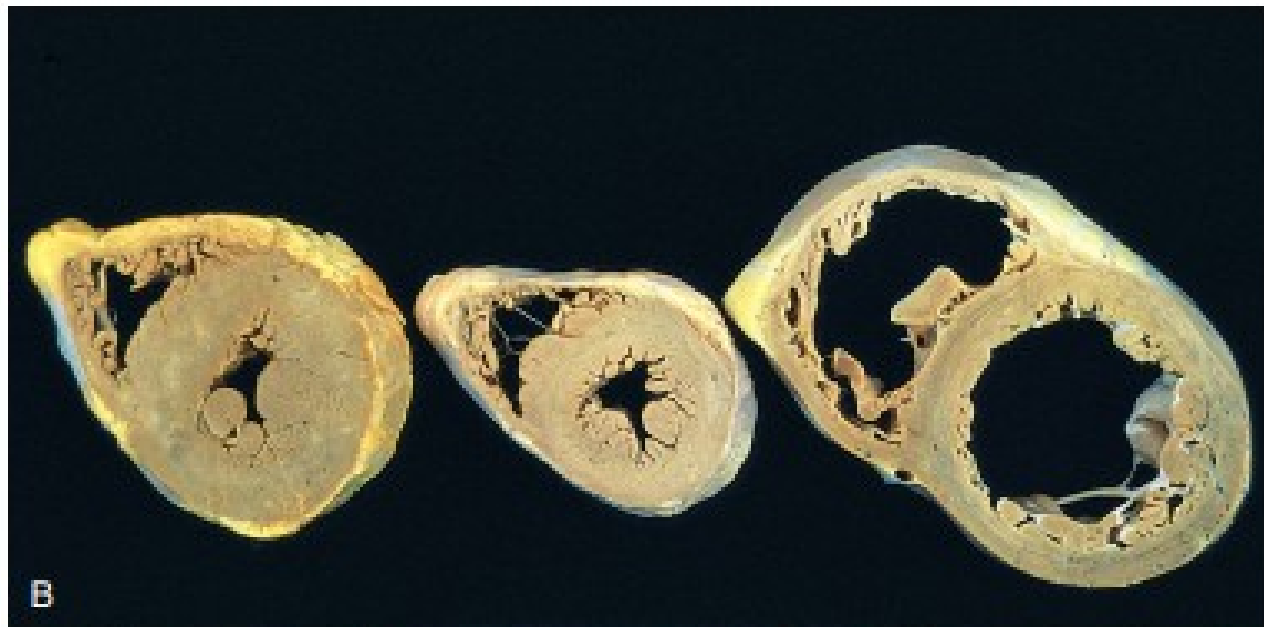
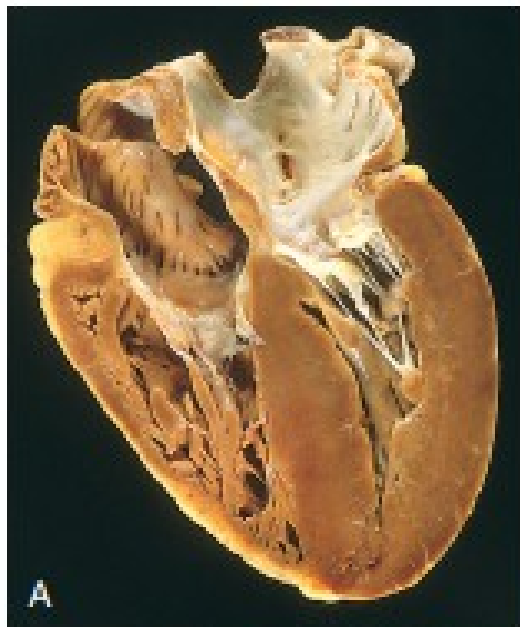
- Increased end-diastolic filling volumes dilate the heart and cause increased cardiac myofiber stretching; these lengthened fibers contract more forcibly, thereby increasing cardiac output.
- Patient is said to be in compensated heart failure.
- With time the patient develops decompensated heart failure



Activation of neurohumoral systems:

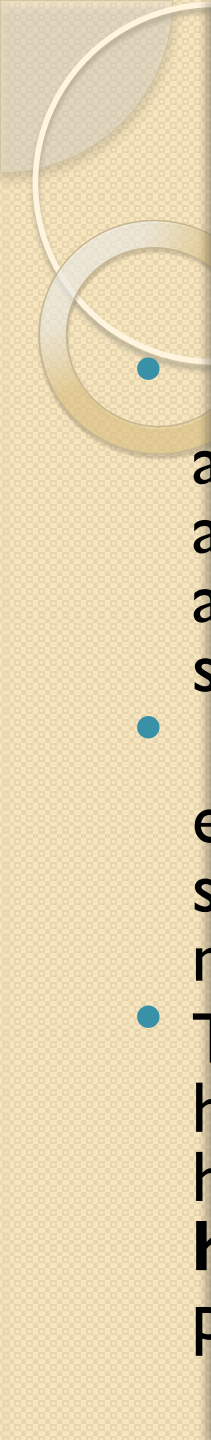
- **Release of the neurotransmitter norepinephrine** by the autonomic nervous system increases heart rate and augments myocardial contractility and vascular resistance.
- **Activation of the renin-angiotensin-aldosterone system** spurs water and salt retention (augmenting circulatory volume) and increases vascular tone.
- **Release of atrial natriuretic peptide** acts to balance the renin-angiotensin-aldosterone system through diuresis and vascular smooth muscle relaxation.

- 
- **Myocardial structural changes**, including augmented muscle mass.
 - Cardiac myocytes adapt to increased workload by assembling new sarcomeres, a change that is accompanied by myocyte enlargement (**hypertrophy**)
 - In pressure overload states concentric hypertrophy occurs
 - In volume overload states ventricle tends to dilate, and the resulting wall thickness can be increased, normal, or decreased (cardiomegally / increased weight)
 - **Compensatory hypertrophy comes at a cost.**
 - The myocardium becomes vulnerable to ischemic injury
 - Cardiac hypertrophy is an independent risk factor for sudden cardiac death.

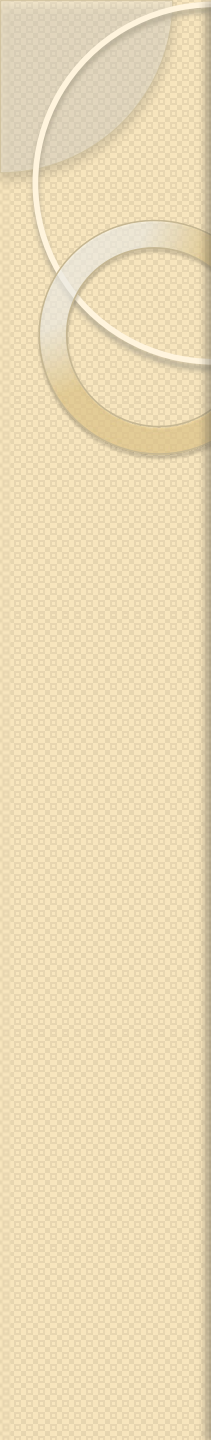


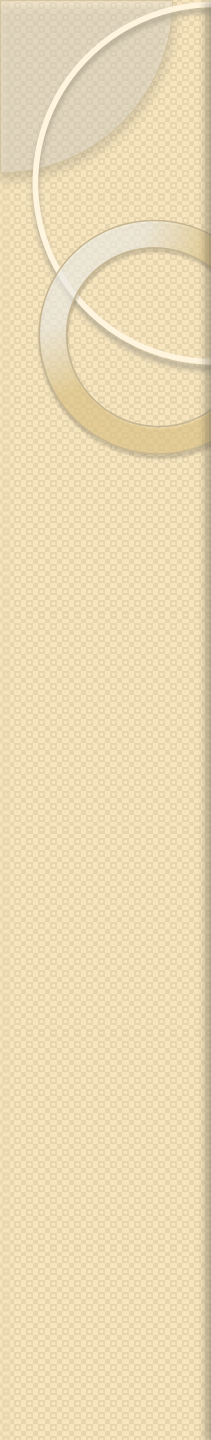
left-sided heart failure

- Heart failure can affect predominantly the left or the right side of the heart or may involve both sides.
- The most common causes of left-sided cardiac failure are
 - ❑ Ischemic heart disease (IHD)
 - ❑ Systemic hypertension
 - ❑ Mitral or aortic valve disease
 - ❑ Primary diseases of the myocardium (e.g., amyloidosis).
- The morphologic and clinical effects of left-sided CHF stem from **diminished systemic perfusion and elevated back-pressures within the pulmonary circulation**

- 
- In acute left side heart failure the lungs are heavy and boggy, and microscopically show perivascular and interstitial transudates, alveolar septal edema, and accumulation of edema fluid in the alveolar spaces.
 - In chronic heart failure, variable numbers of red cells extravasate from the leaky capillaries into alveolar spaces, where they are phagocytosed by macrophages.
 - The subsequent breakdown of red cells and hemoglobin leads to the appearance of hemosiderin-laden alveolar macrophages—so-called **heart failure cells**—that reflect previous episodes of pulmonary edema.

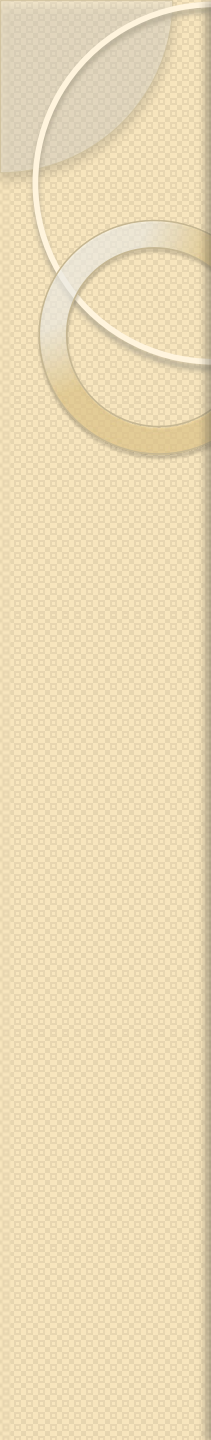
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- **Dyspnea (shortness of breath)** on exertion is usually the earliest and most significant symptom of left-sided heart failure
 - **Cough** is also common as a consequence of fluid transudation into air spaces.
 - As failure progresses, patients experience dyspnea when recumbent (**orthopnea**)
 - This occurs because the supine position increases venous return from the lower extremities and also elevates the diaphragm.
 - Orthopnea typically is relieved by **sitting or standing, so patients usually sleep in a semi-seated position.**
 - Paroxysmal nocturnal dyspnea is a particularly dramatic form of breathlessness, awakening patients from sleep with extreme dyspnea bordering on feelings of suffocation.

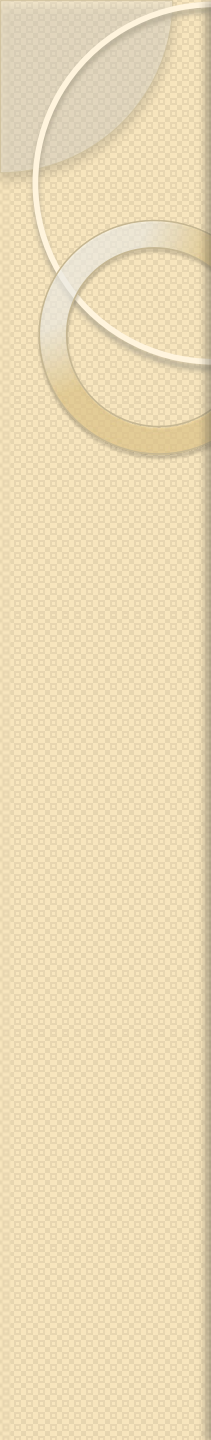
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- Other manifestations of left ventricular failure include an enlarged heart (**cardiomegaly**), **tachycardia**, a **third heart sound (S3)**, and **fine ráles** at the lung bases, caused by the opening of edematous pulmonary alveoli.
 - **Atrial fibrillation** also causes stasis of the blood (particularly in the atrial appendage), frequently leading to the formation of **thrombi** that can shed emboli and cause strokes and manifestations of infarction in other organs.
 - Diminished cardiac output leads to decreased renal perfusion that in turn triggers the renin-angiotensin-aldosterone axis, increasing intravascular volume and pressures

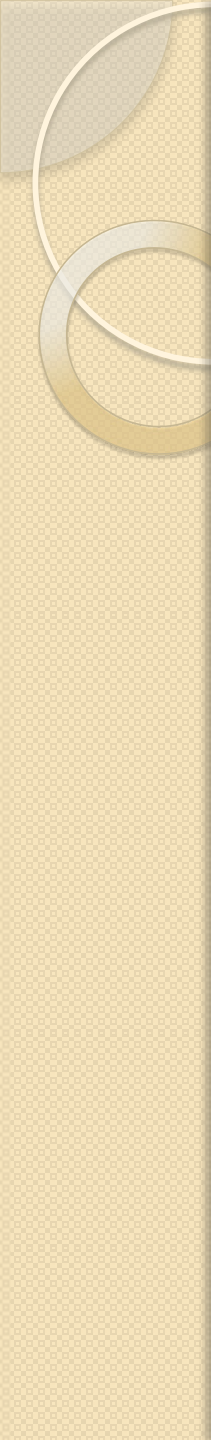
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- Unfortunately, with a failing heart, these compensatory effects exacerbate the pulmonary edema.
 - With further progression of CHF, **prerenal azotemia** may supervene, with impaired excretion of nitrogenous wastes and increasing metabolic derangement.
 - In severe CHF, diminished cerebral perfusion may manifest as hypoxic encephalopathy marked by irritability, diminished cognition, and restlessness that can progress to stupor and coma.

Right-Sided Heart Failure

- Is usually the consequence of left-sided heart failure, since any pressure increase in the pulmonary circulation inevitably produces an increased burden on the right side of the heart.
- Isolated right-sided heart failure is infrequent and typically occurs in patients with one of a variety of disorders affecting the lungs hence it is often referred to as **cor pulmonale**.
- In cor pulmonale, myocardial hypertrophy and dilation generally are confined to the right ventricle and atrium
- In pure right-sided heart failure engorgement of the systemic and portal venous systems typically is pronounced and pulmonary congestion is minimal.

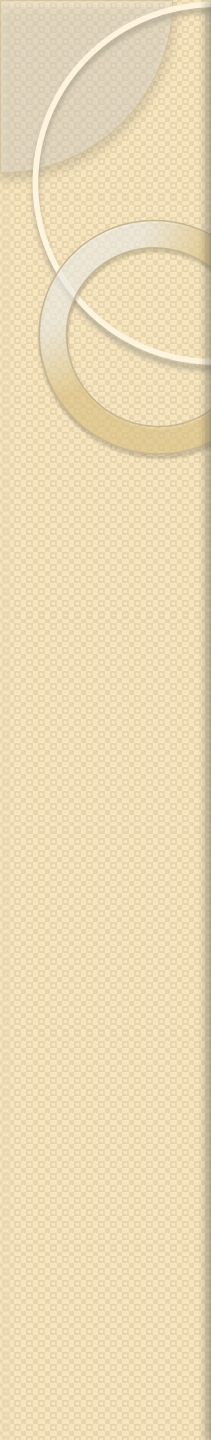
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- The liver usually is increased in size and weight (congestive hepatomegaly)
 - A cut section displays prominent passive congestion, a pattern referred to as **nutmeg liver**
 - With long-standing severe right-sided heart failure, **cardiac cirrhosis develops**.
 - Right-sided heart failure can also lead to elevated pressure in the portal vein and its tributaries (portal hypertension), with vascular congestion producing a tense, enlarged spleen (**congestive splenomegaly**).

- 
- Systemic venous congestion due to right-sided heart failure can lead to transudates (**effusions**) in the pleural and pericardial spaces, but usually does not cause pulmonary parenchymal edema.
 - A combination of hepatic congestion (with or without diminished albumin synthesis) and portal hypertension can lead to peritoneal transudates (ascites).
 - Edema of dependent portions of the body, especially the feet and lower legs, is a hallmark of right sided CHF.

- 
- Pure right-sided heart failure typically is not associated with respiratory symptoms
 - The clinical manifestations are related to systemic and portal venous congestion and include **hepatic and splenic enlargement, peripheral edema, pleural effusion, and ascites**
 - Venous congestion and hypoxia of the kidneys and brain due to right-sided heart failure can produce deficits comparable to those caused by the hypoperfusion of left-sided heart failure.

CONGENITAL HEART DISEASE

- Developmental abnormalities of heart & great arteries present from birth.
- They account for 20% to 30% of all birth defects.
- The incidence is higher in premature infants and in stillborn.
- The number of patients surviving with congenital heart disease is increasing rapidly.
- Congenital heart disease most commonly arises from **faulty embryogenesis** during gestational weeks 3 through 8, when major cardiovascular structures develop
- Incidence **6-8/1 000 live births**

- 
- Cause unknown in >90%
 - Environmental exposures, including **congenital rubella infection, teratogens, and maternal diabetes, and genetic factors** are best characterized accepted causes
 - 2-10X increased incidence in siblings or child of affected parent

Two major categories of structural abnormalities

1. Shunt : abnormal communication

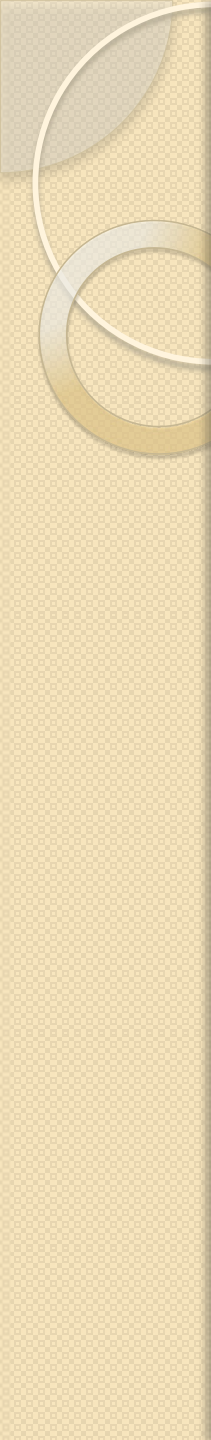
2. Obstruction

Stenosis: partial narrowing

Atresia: complete obstruction

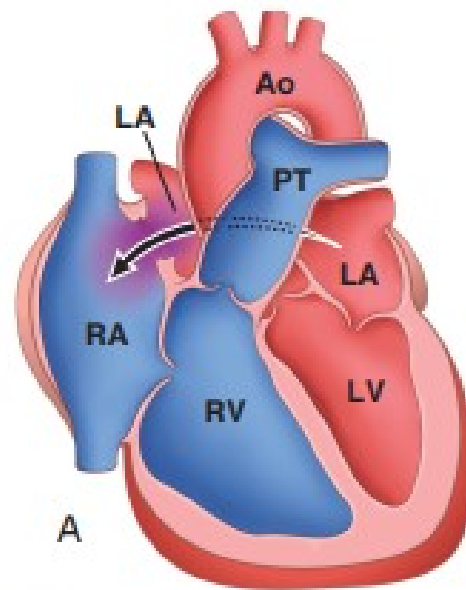
- A shunt is an abnormal communication between chambers or blood vessels.

Malformation	Incidence per 1 Million Live Births	%
Ventricular septal defect	4482	42
Atrial septal defect	1043	10
Pulmonary stenosis	836	8
Patent ductus arteriosus	781	7
Tetralogy of Fallot	577	5
Coarctation of aorta	492	5
Atrioventricular septal defect	396	4
Aortic stenosis	388	4
Transposition of great arteries	388	4
Truncus arteriosus	136	1
Total anomalous pulmonary venous connection	120	1
Tricuspid atresia	118	1
TOTAL	9757	

- 
- Ventricular septal defect (VSD) 30%
 - Patent ductus arteriosus (PDA) 10%
 - Pulmonary stenosis 10%
 - Atrial septal defect (ASD) 7%
 - Tetralogy of Fallot 5%
 - Coarctation of aorta 5%
 - Aortic stenosis 5%

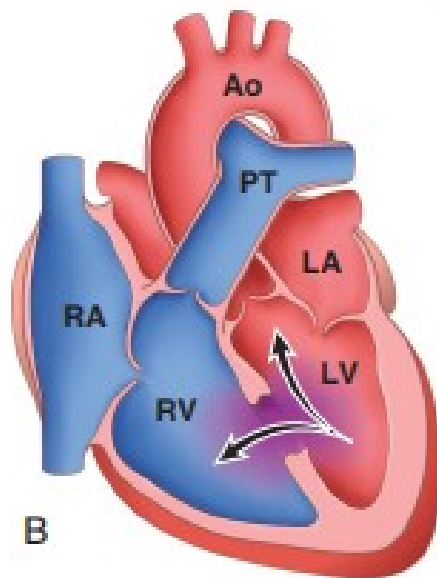
Malformations Associated With Left-to-Right Shunts

- Are the most common types of congenital cardiac malformations.
- They include atrial septal defects (ASDs), ventricular septal defects (VSDs), and patent ductus arteriosus (PDA)
- ASDs typically increase only right ventricular and pulmonary outflow volumes.
- VSDs and PDAs cause both increased pulmonary blood flow and pressure.
- Manifestations of these shunts range from completely asymptomatic to fulminant heart failure.
- Cyanosis is not an early feature of these defects.



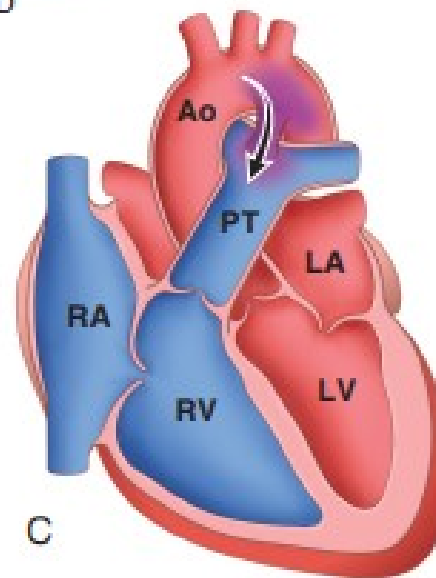
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ASD



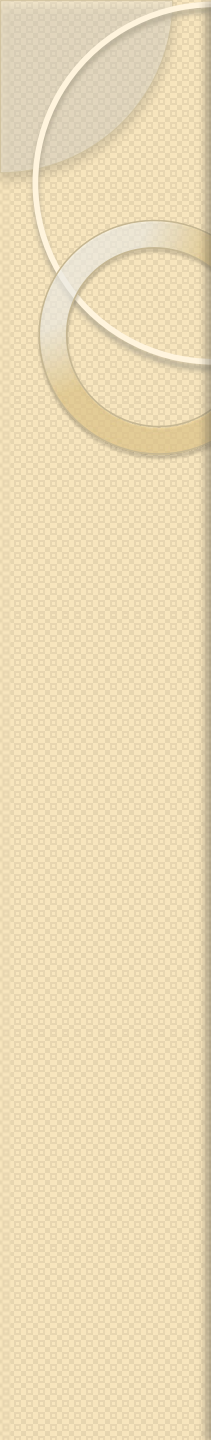
B

VSD



C

PDA

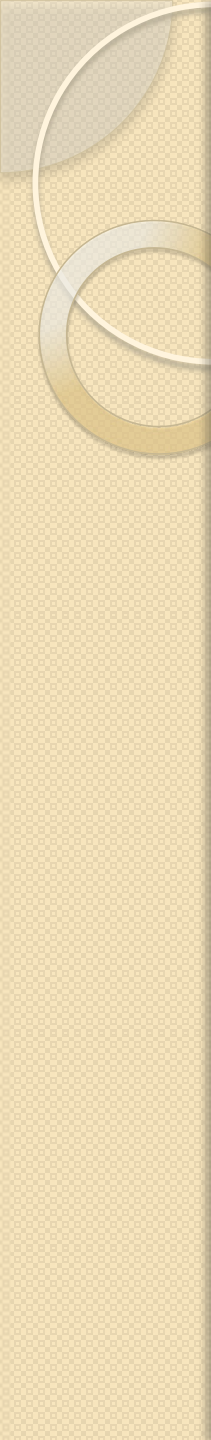
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- shunting may eventually give rise to pulmonary hypertension and right-to-left shunting of unoxygenated blood into the systemic circulation, a change marked by the appearance of cyanosis **(Eisenmenger syndrome)**.
 - Once significant pulmonary hypertension develops, the structural defects of congenital heart disease are considered irreversible.

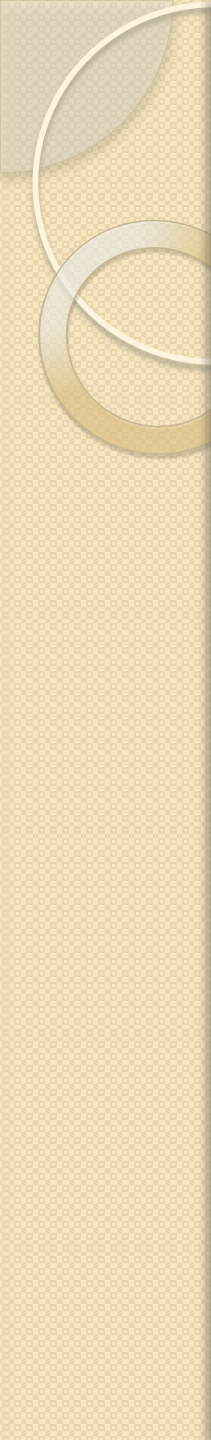
Atrial Septal Defect (ASD)

- Abnormal communication in atrial septum (AS) allows blood flow between LA & RA
- ASDs usually are asymptomatic until adulthood
- The most common defects to be first diagnosed in adults.
- Classified by size & location in AS
 - ASD at fossa ovalis (secundum) 90%
 - Primum ASD (low in septum) 5%
- Functional result
 - Left to right shunt: acyanotic
 - Increased pulmonary flow
- Systolic murmur
- May be diagnosed in adults > 30 yrs old (low pressure, low flow)
- **Less than 10% develop pulmonary hypertension**
- Surgical or intravascular ASD closure is thus performed to prevent the development of heart failure, paradoxical embolization, and irreversible pulmonary vascular disease.

Ventricular Septal Defect (VSD)

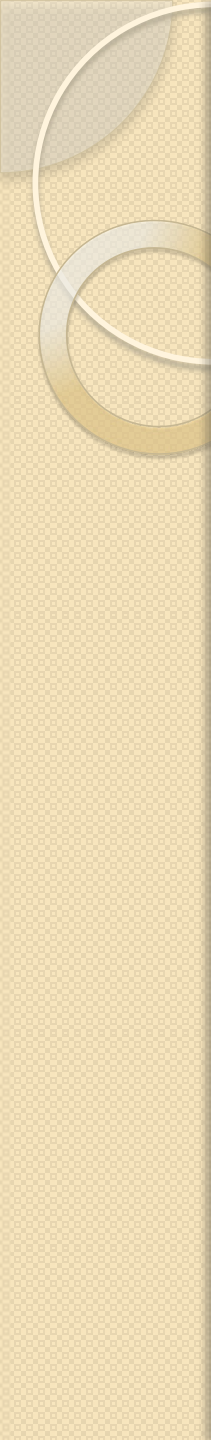
- Defect in ventricular septum allows flow between left and right left ventricle
- The most common congenital cardiac anomaly at birth
- Functional significance depends on size of defect (therefore, with resistance determines amount of flow across defect)
- The basal (membranous) region is the last part of the septum to develop and is the site of approximately 90% of VSDs.
- Most VSDs close spontaneously in childhood, but only 20% to 30% of VSDs occur in isolation; the remainder are associated with other cardiac malformations.
- L to R shunt is acyanotic (increased pulmonary flow); high pressure, high flow
- Pan-systolic murmur
- 50% close spontaneously in 1st year

- 
- In defects associated with a significant left-to-right shunt:
 - ❑ The right ventricle is hypertrophied and often dilated.
 - ❑ The diameter of the pulmonary artery is increased, owing to the increased right ventricular output and higher right sided pressures.
 - ❑ Vascular changes typical of pulmonary hypertension are common

- 
- Larger defects result in chronic left-to-right shunting, often complicated by pulmonary hypertension and CHF.
 - Progressive pulmonary hypertension, with resultant reversal of the shunt and cyanosis, occurs earlier and more frequently with VSDs than with ASDs.
 - Early surgical correction is therefore indicated for such lesions

Patent Ductus Arteriosus

- The ductus arteriosus arises from the left pulmonary artery and joins the aorta just distal to the origin of the left subclavian artery.
- During intrauterine life, it permits blood to flow from the pulmonary artery to the aorta, bypassing the unoxygenated lungs.
- Within 1 to 2 days of birth in healthy term infants, the ductus constricts and closes
- Ductal closure can be delayed (or even absent) in infants with hypoxia (related to respiratory distress or heart disease).
- PDAs account for about 7% of congenital heart lesions ,the great majority of these (90%) are isolated defects.

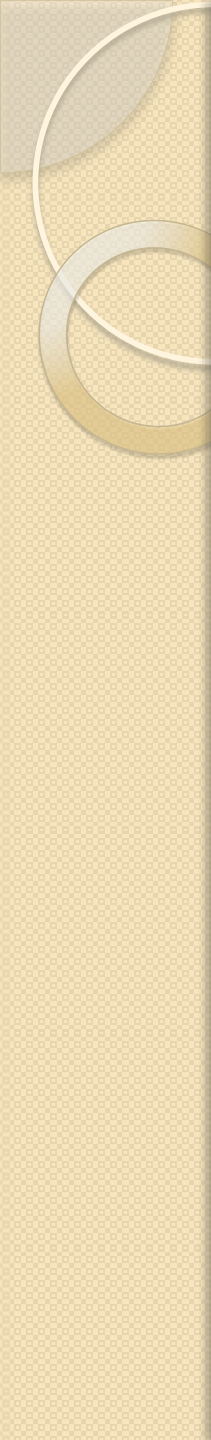
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- PDAs are high-pressure left-to-right shunts that produce **harsh, “machinery-like” murmurs**.
 - A small PDA generally causes no symptoms, although larger defects eventually can lead to Eisenmenger syndrome with cyanosis and congestive heart failure.
 - High-pressure shunts also predispose patients to developing infective endocarditis.
 - While isolated PDAs should be closed as early in life as is feasible, preservation of ductal patency (by administering prostaglandin E) can be life saving when a PDA is the only means to sustain systemic or pulmonary blood flow (e.g., in infants with aortic or pulmonic atresia)

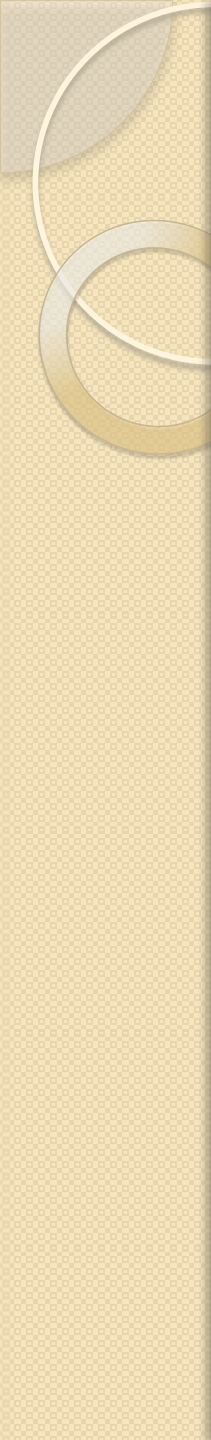
Malformations Associated With Right-to-Left Shunts

- Cardiac malformations associated with right-to-left shunts are distinguished by early cyanosis.
- Two of the most important are **tetralogy of Fallot** and **transposition of the great vessels**
- Clinical consequences of severe, systemic cyanosis include clubbing of the tips of the fingers and **toes (hypertrophic osteoarthropathy)** polycythemia, and **paradoxical embolization**

Tetralogy of Fallot (TOF)

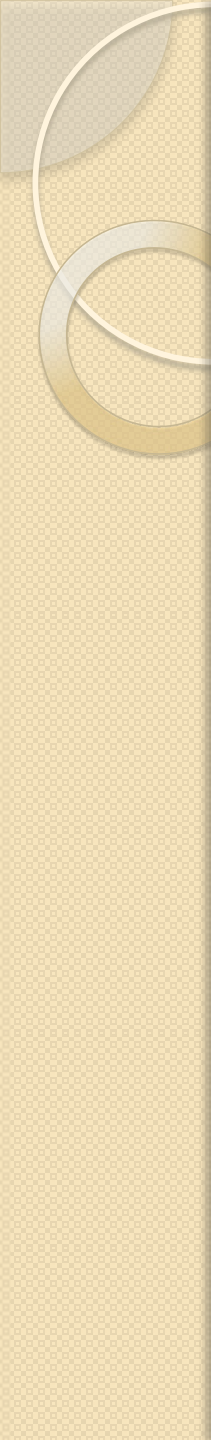
- Most common form of ***cyanotic*** heart disease
- 4 components
 - VSD
 - Right ventricular outflow tract obstruction (subpulmonic stenosis)
 - Overriding of the VSD by the aorta
 - RV hypertrophy (due to PS)
- Functional consequences
 - Right to left shunt
 - Decreased pulmonary flow
- Clinical murmur

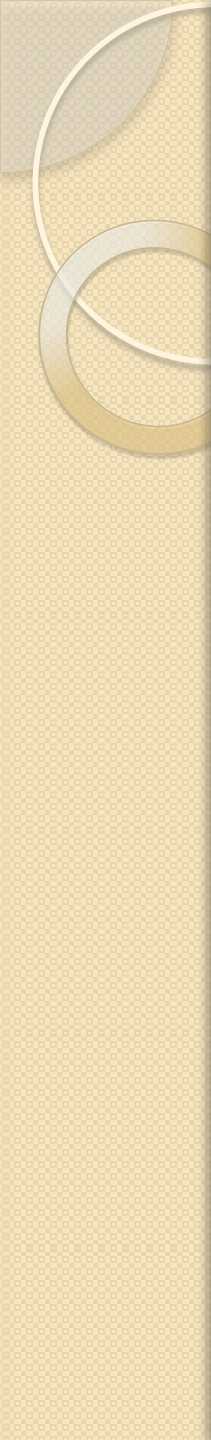
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- The heart is enlarged and “boot-shaped” as a consequence of right ventricular hypertrophy; the proximal aorta is dilated, while the pulmonary trunk is hypoplastic.
 - The VSD usually is large
 - The hemodynamic consequences of tetralogy of Fallot are right-to-left shunting, decreased pulmonary blood flow, and increased aortic volumes
 - The clinical severity largely depends on the degree of the **pulmonary outflow obstruction**

- 
- More commonly, more severe degrees of pulmonic stenosis cause early cyanosis
 - patients develop the typical sequelae of cyanotic heart disease, such as hypertrophic osteoarthropathy and polycythemia (due to hypoxia)
 - with attendant hyper viscosity; right-to-left shunting also increases the risk for infective endocarditis and systemic embolization.
 - Complete surgical repair is possible with classic tetralogy of Fallot but is more complicated in the setting of pulmonary atresia.

Transposition of the Great Arteries

- The aorta arises from the right ventricle and the pulmonary artery emanates from the left ventricle
- The functional outcome is separation of the systemic and pulmonary circulations, a condition incompatible with postnatal life unless a shunt (such as a VSD) allows delivery of oxygenated blood to the aorta
- Indeed, VSDs occur in one third of cases

- 
- Some newborns with transposition of the great arteries have a patent foramen ovale or PDA that allows oxygenated blood to reach the aorta
 - But these tend to close; such infants typically require emergent surgical intervention within the first few days of life.
 - Most patients with uncorrected transposition of the great arteries die within the first months of life.

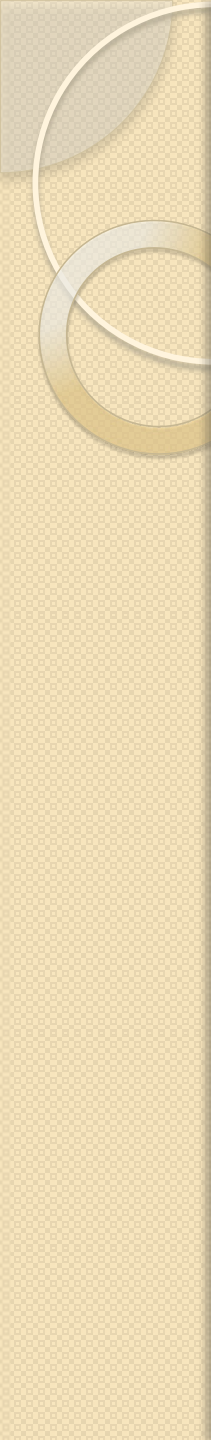


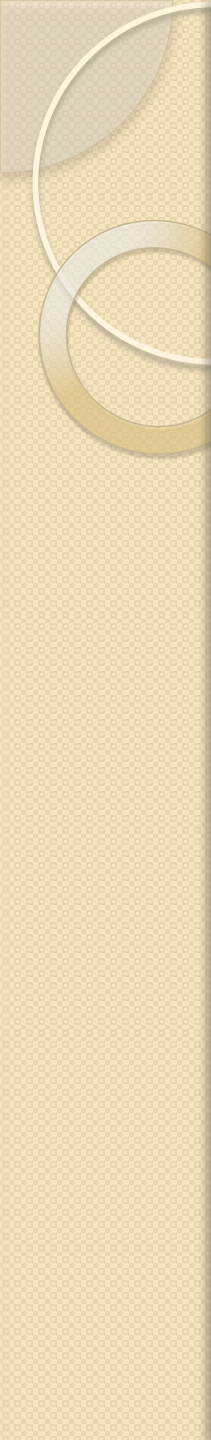
Malformations Associated With Obstructive Lesions

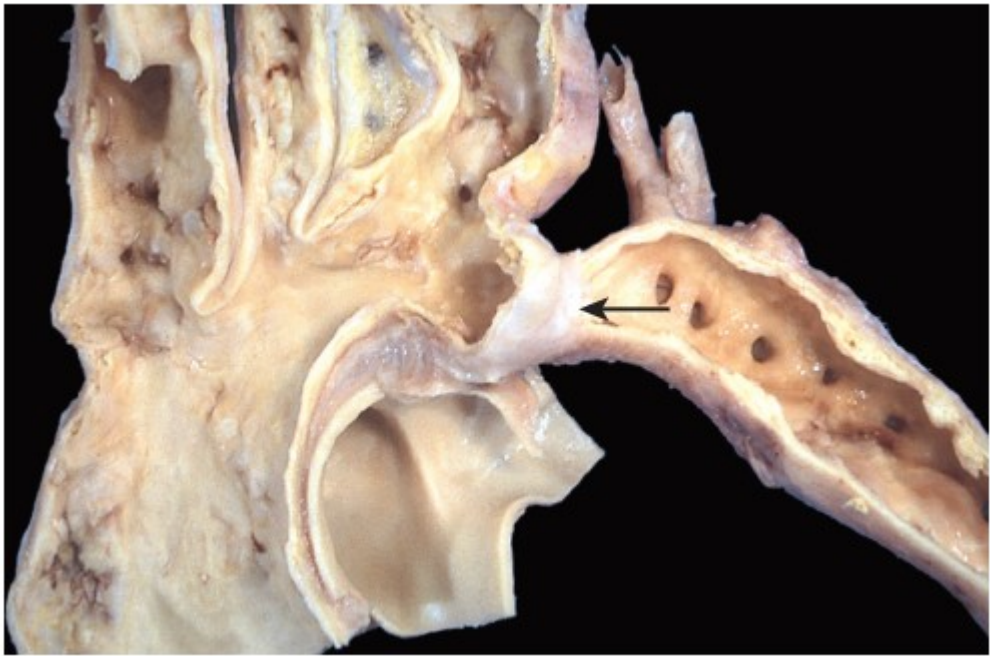
- Relatively common examples of congenital obstructions are pulmonic valve stenosis, aortic valve stenosis or atresia, and coarctation of the aorta.

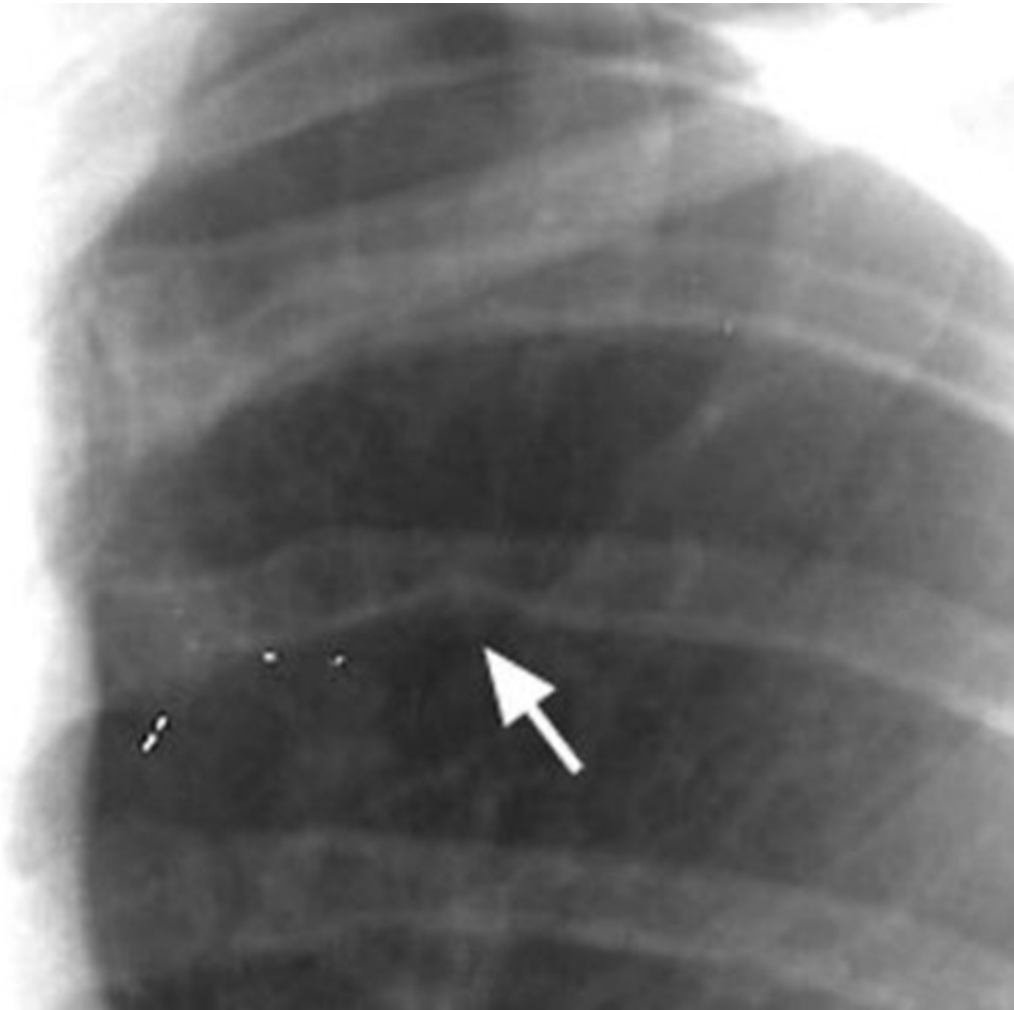
Aortic Coarctation

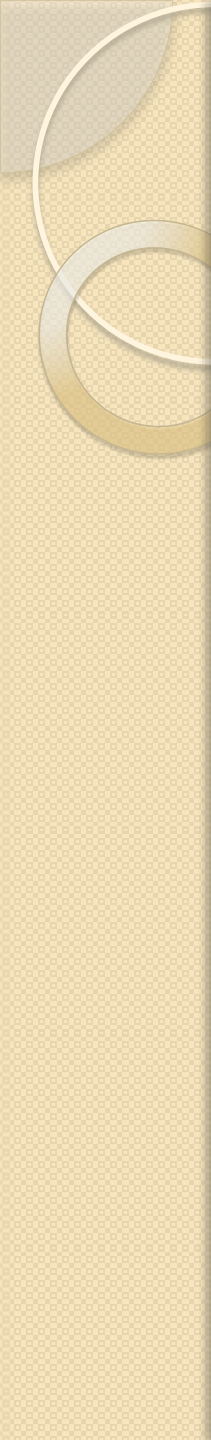
- Coarctation (narrowing, or constriction) of the aorta is a common form of obstructive congenital heart disease.
- Males are affected twice as often as females, although females with Turner syndrome frequently have coarctation
- Localized obstruction due to curtain-like fold of aortic media that narrows lumen
- There are two classic forms :
 - An “infantile” preductal form featuring hypoplasia of the aortic arch proximal to a PDA
 - An “adult” postductal form consisting of a discrete ridge like infolding of the aorta, adjacent to the ligamentum arteriosum
- Often sited just beyond origin of LSA
- May be associated with other CV abnormalities
- May give symptoms in infancy
- Lower blood pressure in legs than arms

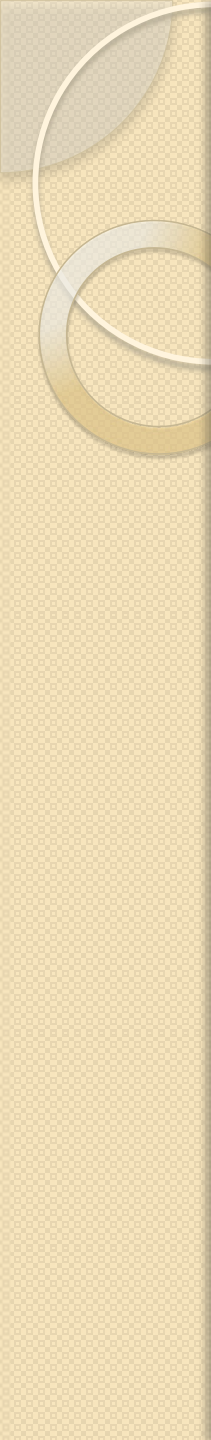
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- Clinical manifestations depend on the severity of the narrowing and the patency of the ductus arteriosus
 - Preductal coarctation with a PDA usually presents early in life, classically as cyanosis localized to the lower half of the body; without intervention, most affected infants die in the neonatal period.
 - Postductal coarctation without a PDA usually is asymptomatic, and the disease may remain unrecognized well into adult life.
 - Classically, there is upper-extremity hypertension paired with weak pulses and relative hypotension in the lower extremities, associated with symptoms of claudication and coldness

- 
- Exuberant collateral circulation “around” the coarctation often develops through markedly enlarged **intercostal and internal mammary arteries**;
 - Expansion of the flow through these vessels can lead to radiographically visible **“notching” of the ribs.**





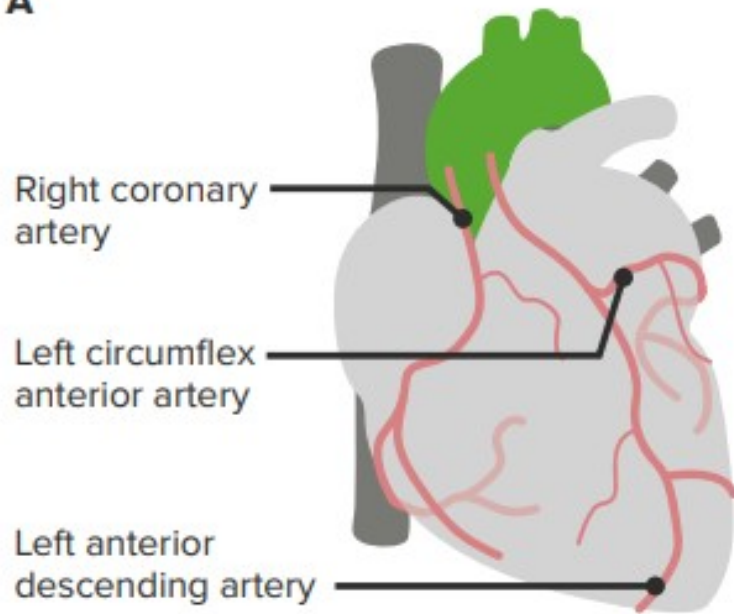
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- **Diagnosis**
 - ECG
 - CXR (Posterior rib notching of 3rd-8th ribs)
 - Echocardiography
 - **Treatment**
 - Balloon angioplasty w/ stenting
 - Surgery
 - **Prognosis**
 - Mean survival if untreated is 35 years



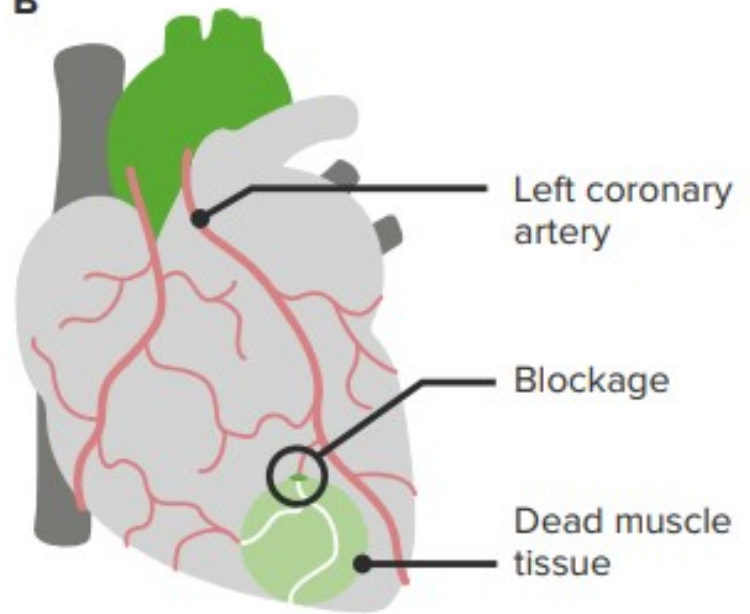
Ischemic Heart Diseases

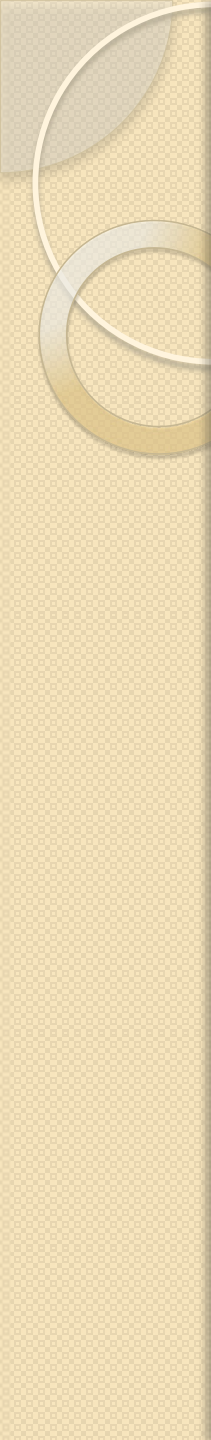
- Ischemic Heart Disease (IHD), also termed Coronary Artery Disease (CAD), are terms used to describe a wide range of clinical conditions in which an imbalance in oxygen supply and myocardial demand results in partial ischemia of the myocardium.

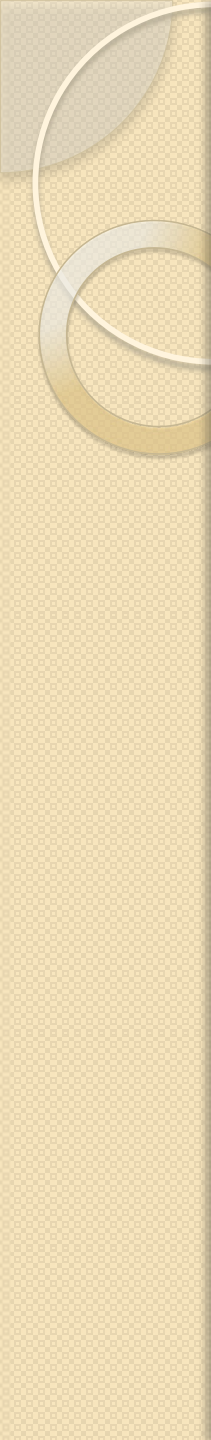
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B



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- The most common risk factor of coronary heart disease is **atherosclerosis** of the epicardial coronary arteries, resulting in partial or complete obstruction with subsequent inadequate perfusion of the area of the myocardium supplied by the involved coronary artery.
 - Main risk factors (in addition to increased LDL or low HDL cholesterol) are **arterial hypertension, diabetes mellitus, smoking, a positive family history, and male gender**

- 
- 4 major factors determine the oxygen demand of the myocardium:

1. Heart rate

2. Systolic blood pressure (afterload)

3. Tension on myocardial wall (preload)

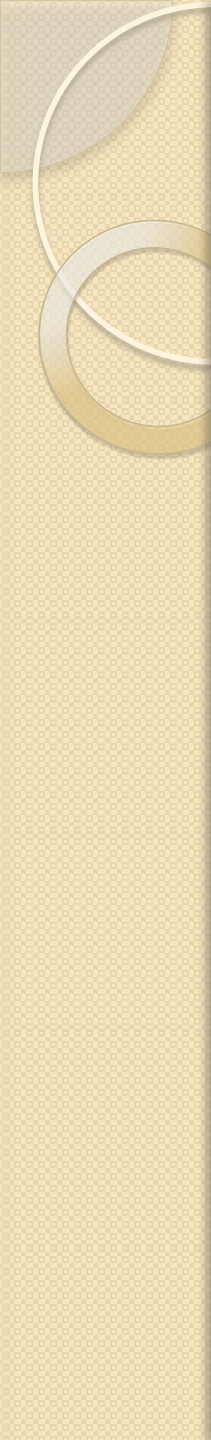
4. Myocardial contractility

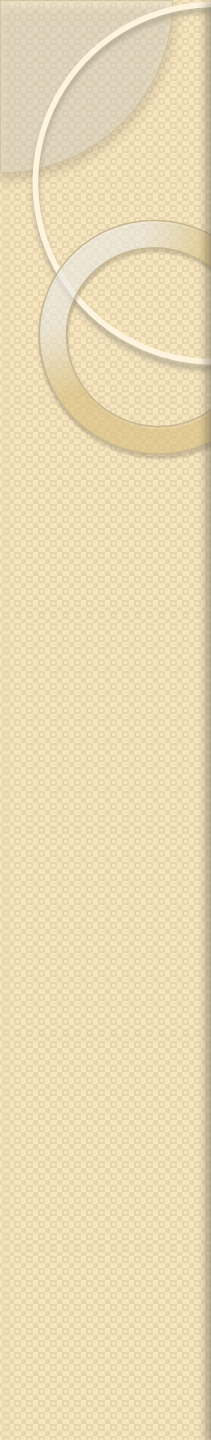
- Any clinical condition which increases these factors will also increase myocardial oxygen demand and can result in ischemia.
- These conditions include **extreme tachycardia, hypertension, and ventricular hypertrophy**

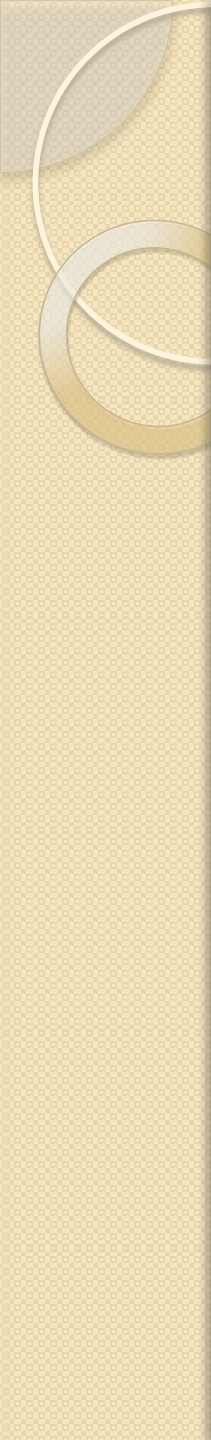
Classification of IHD

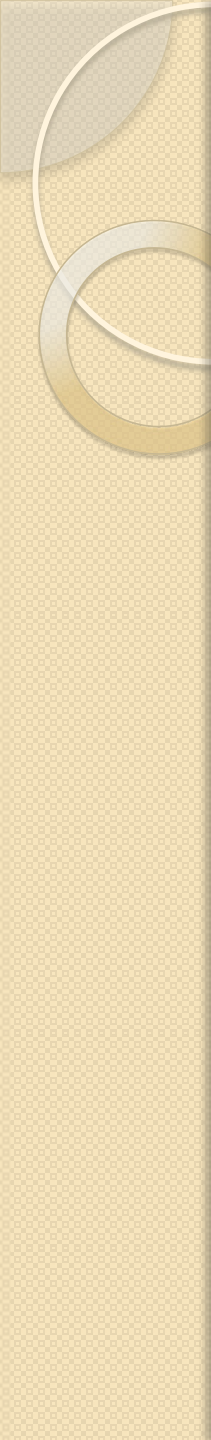
- Patients with coronary heart disease (IHD) can present with either:
 1. Chronic artery disease (CAD), which most commonly presents as stable angina.
 2. Acute coronary syndromes (ACSs), which is a term that encompasses:
 - Unstable Angina (UA)
 - Myocardial infarction, divided into:
 - ❑ ST-segment elevation myocardial infarction (STEMI)
 - ❑ Non-ST-segment elevation myocardial infarction (NSTEMI)

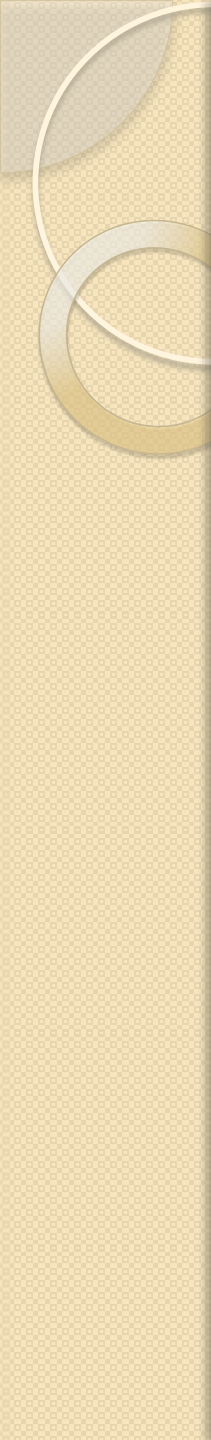
- **Variant angina**
- Variant angina is caused by a vasospasm which occurs at rest.
- **Prognosis of IHD**
- Several prognostic indicators determine the outcome of coronary heart disease(IHD):
 - ❑ Function of the left ventricle: Increased left ventricular end-diastolic pressure, increased ventricular volume and reduced ejection fraction are associated with poor prognosis.
 - ❑ Location and severity of coronary artery stenosis: Stenosis of the main left anterior descending coronary artery is associated with greater risk and poor prognosis.
 - ❑ Number and severity of risk factors: A large number of risk factors for atherosclerosis are associated with increased risk of myocardial infarction with a worse prognosis

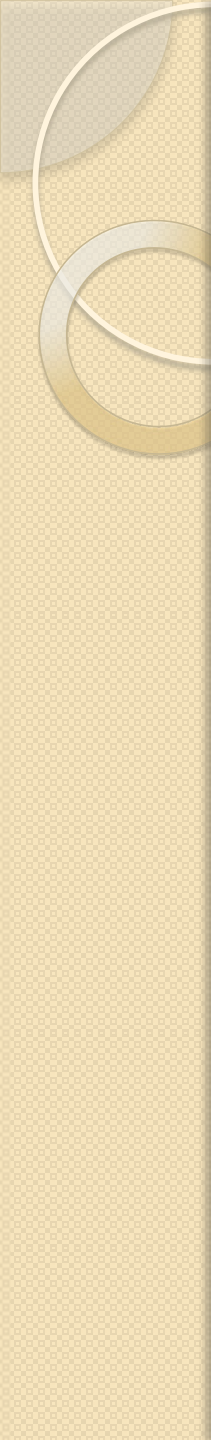
- 
- **Stable angina** is defined as chest pain repeatedly triggered by specific events, such as physical exertion, psychological stress, and exposure to the cold.
 - It usually subsides within < 20 minutes of rest or after administration of nitrates.
 - The typical patient presents with episodes of chest discomfort described as a sense of pressure, choking, heaviness, or tightness in the chest.

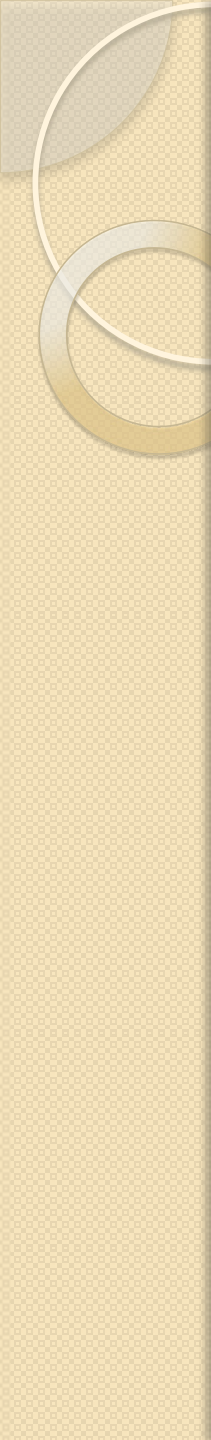
- 
- The pain starts gradually, with the intensity increasing and decreasing (crescendo decrescendo in nature) within minutes and typically lasting for 2–5 minutes.
 - It generally does not last for 20–30 minutes, unless the patient has acute coronary syndrome.
 - Coronary pain is usually described as central sub sternal discomfort in which patient can't localize the site of pain and typically places a hand or clenched fist over the sternum.

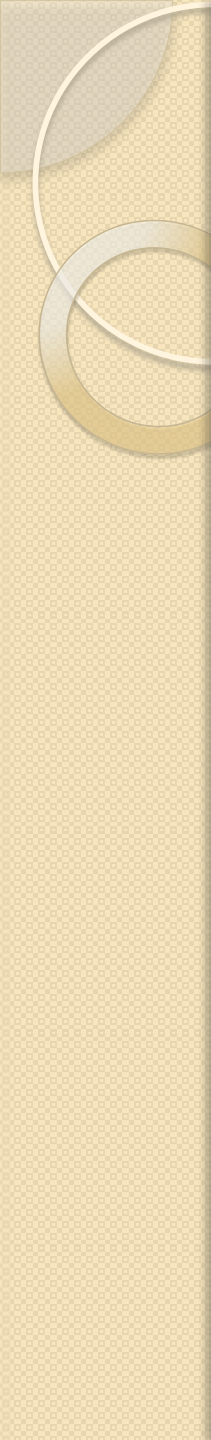
- 
- The pain radiates to any dermatome from C8 to T4, most often to the left shoulder and left arm (especially the ulnar surface).
 - It can also radiate to the inter scapular region, back, epigastrium, and lower jaw.
 - Episodes of angina are provoked by physical exertion and intense emotion and relieved within minutes by **rest and sublingual nitroglycerin.**

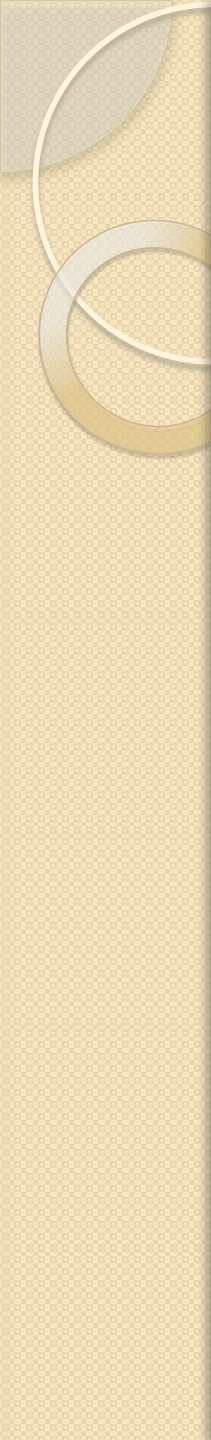
- 
- Angina is usually associated with **shortness of breath, diaphoresis, dizziness, lightheadedness, and fatigue**
 - During the pain, reversible ST-segment depression (injury pattern) or elevation, with or without T wave inversion, is suggestive of myocardial ischemia.

- 
- A sudden coronary artery spasm that leads to a reduction in coronary blood flow, thus causing severe chest pain (angina) at rest, is called **vasospastic, Prinzmetal's, or variant angina**.
 - This type of angina occurs at rest, rather than on exertion, without any initiating factors.
 - This chest pain can **last less than 15–30 minutes**
 - In this angina type, myocardial ischemia is due to transient vasospasm with or without any underlying pathology.

- 
- Patients are generally younger than those with stable or unstable angina secondary to coronary artery atherosclerosis.
 - Variant angina more commonly affects women
 - various risk factors such as **stress, smoking, cocaine use, insulin resistance, and medications** that have the effect of constricting blood vessels, such as triptans.
 - The exact mechanisms that cause coronary artery spasm are not yet well understood.
 - Some proposed theories relate to endothelial dysfunction and increased contractility of vascular smooth muscle

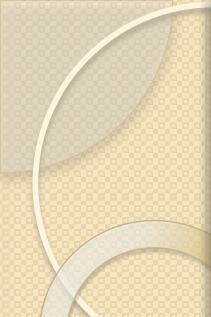
- 
- Patients experience **very severe central chest pain, which is the same type of pain as classic angina.**
 - These attacks tend to happen at rest or during normal activity
 - Other symptoms include **shortness of breath and palpitations.**
 - Patients typically experience attacks of angina in clusters, from **midnight to early morning (midnight – 8:00 am).**
 - Syncope may also occur if there are disturbances to the heart rhythm such as asystole, atrioventricular block, or ventricular tachyarrhythmias.
 - Fatal arrhythmias and sudden cardiac death may also occur

- 
- Electrocardiography is the key to diagnosing vasospastic angina.
 - ECG changes demonstrate transient ST-segment elevation during periods of chest pain, which resolves when the pain settles.
 - ST-segment elevation represents transmural myocardial ischemia and is accompanied by reciprocal ST depressions.
 - arrhythmias may be detectable during an episode of variant angina, such as ventricular tachycardia, atrioventricular block, and bradyarrhythmia

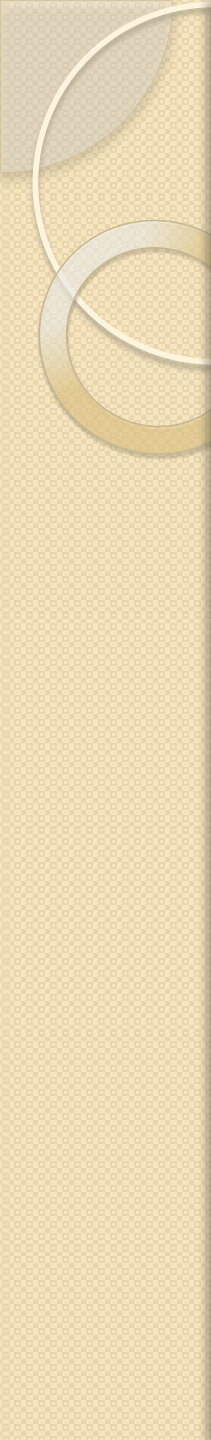
- 
- A nitrate can be administered to relieve an attack of angina in an acute setting.
 - The main drug used to treat variant coronary syndrome is a **calcium channel blocker, such as diltiazem.**
 - Calcium channel blockers are very effective in the prevention of ischemia.
 - They may be given alone or together with isosorbide mononitrate, a long-acting nitrate.
 - Patients who have coronary spasm in multiple arteries are prone to fatal arrhythmias.
 - Beta-blockers are contraindicated as they reduce the dilatation of the smooth musculature through blocking the beta-2 receptors and increase the tone of the coronary vessels.

Acute coronary syndrome

- Acute coronary syndrome (ACS) is characterized by **sudden onset severe chest pain due to partial or complete occlusion of a coronary artery resulting from unstable ruptured plaque in the setting of advanced ischemic heart disease.**
- ACS covers 3 clinical entities:
 - ❑ Unstable angina pectoris
 - ❑ Myocardial infarction, which includes:
 - Non-ST-segment elevation myocardial infarction (NSTEMI)
 - ST-segment elevation myocardial infarction (STEMI)



Diagnosis	Clinical features	ECG findings	Laboratory findings
Unstable angina	Ischemic chest pain that occurs at rest or with previously tolerated levels of exertion	None, or ST-segment depressions	None
Non-ST-elevation Myocardial infarction (NSTEMI)	Ischemic chest pain in any setting	None or ST-segment depressions	Elevated troponin
ST-elevation Myocardial infarction (STEMI)	Ischemic chest pain in any setting	ST-segment elevations	Elevated troponin

- 
- In about **95 % of the cases, a ruptured plaque**, formed inside the coronary arteries within the context of atherosclerosis, is responsible for the symptoms of acute coronary syndrome
 - In perfusion of less than 25 % of the normal flow, the tissue is irreversibly damaged.
 - After 6–12 hours, this damage can be seen by light microscope in the form of **yellowish spots**.

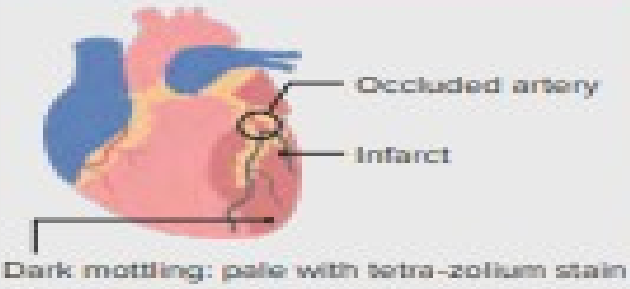
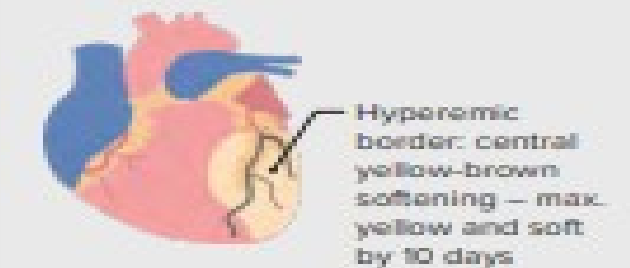
Time	Gross	Light microscope
0–24 hours	 Occluded artery Infarct Dark mottling: pale with tetra-zolium stain	 Early coagulative necrosis, release of necrotic cell contents in blood; edema, hemorrhage, wavy fibers. Neutrophils appear. Reperfusion injury, associated with generation of free radicals, lead to hyperconcentration of myofibrils through $\uparrow$ free calcium influx.
1–3 days	 Hyperemia	 Extensive coagulative necrosis. Tissue surrounding infarct shows acute inflammation with neutrophils.
3–14 days	 Hyperemic border: central yellow-brown softening – max. yellow and soft by 10 days	 Macrophages, then granulation tissue at margins.
2 weeks to 7 months	 Recanalized artery Gray-white	 Contracted scar complete.

Fig. 4-14: Temporal evolution of pathological changes.



Transmural Infarcts



Permanent occlusion of left anterior descending branch



Permanent occlusion of left circumflex branch



Permanent occlusion of right coronary artery (or its posterior descending branch)

Nontransmural Infarcts



Transient/partial obstruction → regional subendocardial infarct



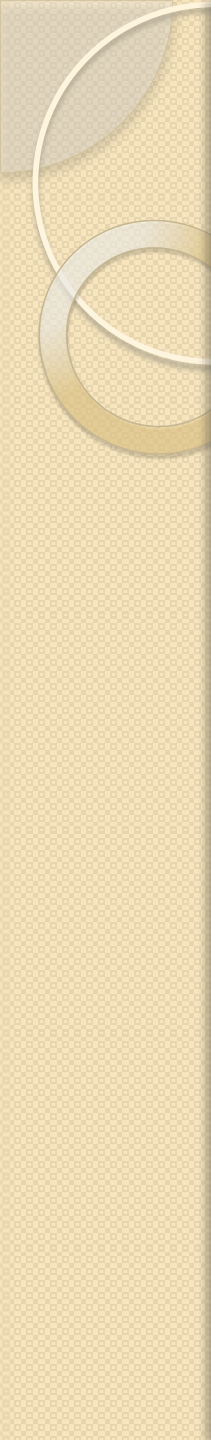
Global hypotension → circumferential subendocardial infarct



Small intramural vessel occlusions → microinfarcts

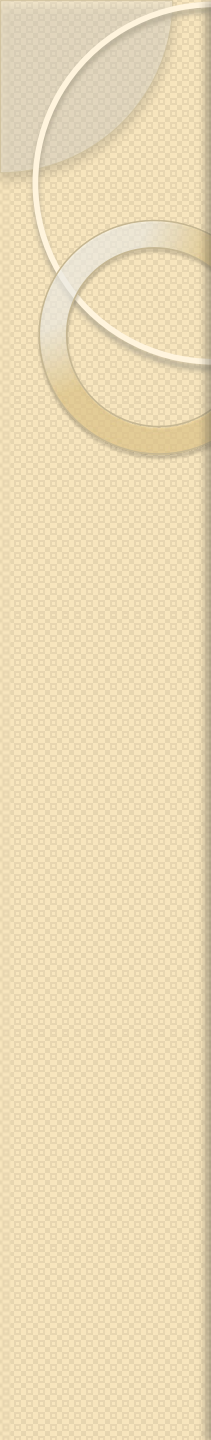
Clinical Features of ACS

- There is often a sudden, severe pain, which typically radiates into the left shoulder and left arm.
- This often occurs without previous exertion, improves only slightly or not at all by administration of nitroglycerin or rest, and lasts longer than 15 minutes.
- Assessment of cardiac enzymes (CK, CK-MB, and troponin) is important to differentiate between different entities of ACS.

- 
- Unstable angina (UA) is characterized by a lack of increase in troponin levels.
 - NSTEMI and STEMI are characterized by destruction of cardiac muscle tissue, which results in elevated cardiac enzymes in the blood
 - Determination of **troponin I or T is considered the gold standard.**
 - Troponin as a cardiac marker is specific to myocardial tissue.
 - Troponin increases shortly after an infarction.
 - About 70 % of patients show an increase 3 hours after the incident and that number rises to 90 % 6 hours after the incident.
 - This timeline stresses the necessity of measuring cardiac enzymes again at a later time, even when the initial result was negative.

The following table provides an overview of the various parameters:

Parameter	Begin of increase	Maximum	Normal levels
Troponin	3–8 hours	12 hours	1–2 weeks
CK-MB	3–12 hours	24 hours	2–3 days
Myoglobin	1–4 hours	4 hours	1 day
LDH	6–12 hours	2–4 days	1–2 weeks
GOT	4 hours	48 hours	3–6 days

- 
- In UA and NSTEMI
 - ECG can be completely normal with no specific ECG changes.
 - ECG may show ST-segment depression or deep-negative T waves.
 - Considered the gold-standard in the diagnosis of ACS because it assesses the exact location and extent of coronary vessel stenosis before possible PCI/surgery

Complications of ACS

- **Early complications**

1. Sudden cardiac death.
2. AV block and ventricular tachyarrhythmias, which could be fatal.
3. Acute left heart failure (pulmonary edema) that results from acute pump failure of heart and acute mitral valve insufficiency secondary to papillary muscle dysfunction.
4. Cardiogenic shock is the most serious complication, with mortality rates exceeding 50 %

- **Late complications**

1. Pericarditis – further classified into:

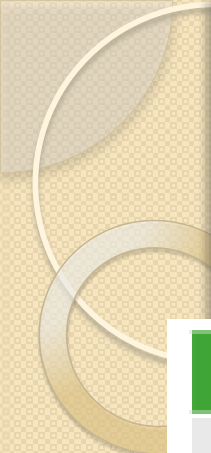
- Early infarct-associated pericarditis which occurs within the first week of MI
- Dressler syndrome, a pericarditis that occurs 2–20 weeks post-MI without an infective cause secondary to circulating antibodies against cardiac muscle cells

2. Congestive heart failure.

3. Re-infarction due to an unstable plaque.

4. Atrial and ventricular aneurysms which may precipitate:

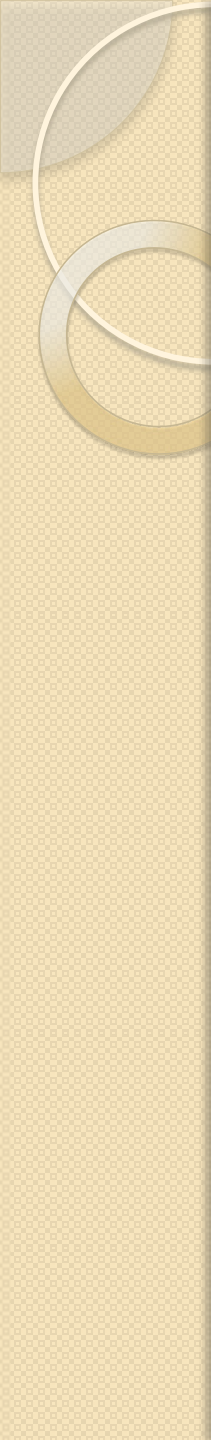
- Thrombus formation and increased risk of thromboembolism
- Cardiac arrhythmia
- Rupture leading to pericardial effusion and cardiac tamponade

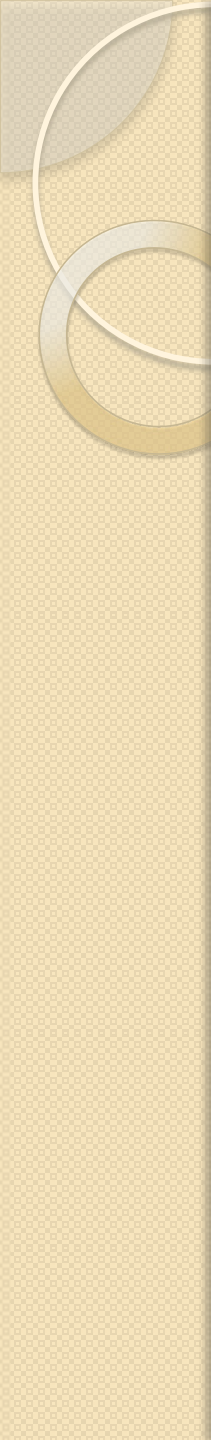


Time	Complications
0–24 h	Ventricular arrhythmia, HF, cardiogenic shock
1–3 days	Postinfarction fibrinous pericarditis
3–14 days	Free wall rupture → Tamponade, Papillary muscle rupture → Mitral regurgitation, Interventricular septal rupture due to macrophage-mediated structural degradation, LV pseudoaneurysm (risk of rupture)
2 weeks to several months	Dressler syndrome, HF, arrhythmias, true ventricular aneurysm (risk of mural thrombus)

Unstable Angina

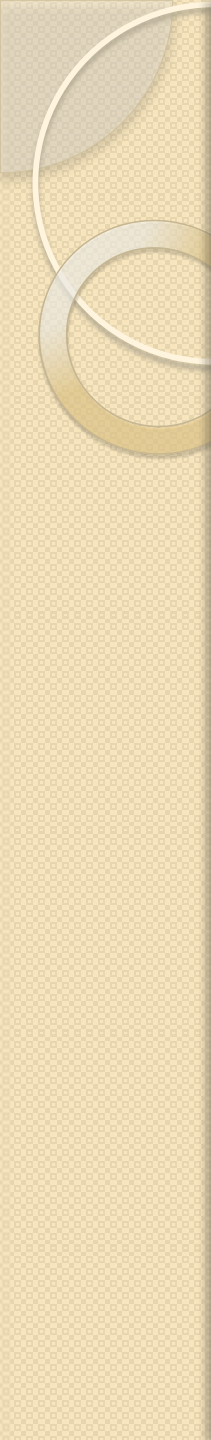
- Unstable angina refers to chest pain that persists longer than 20 minutes, is of increasing intensity, and occurs even at rest.
- Unstable angina (UA) is characterized by the absence of myocardial death, in contrast to non-ST-elevation MI (NSTEMI) which presents with evidence of myocardial necrosis.

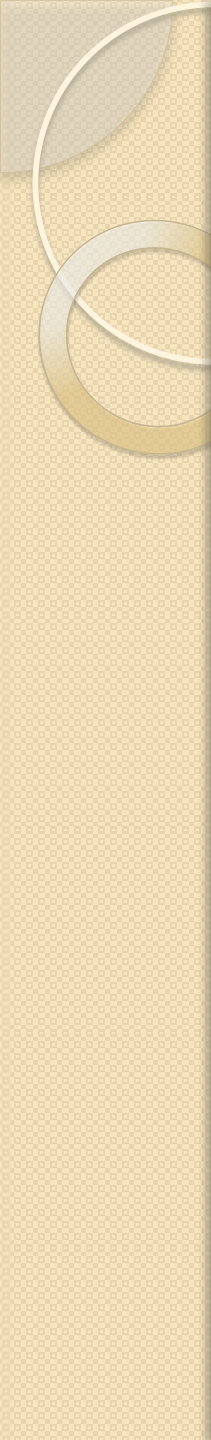
- 
- Transient or persistent ST-segment depression and/or T wave inversion in 0–50 % of patients
 - ECG can be normal



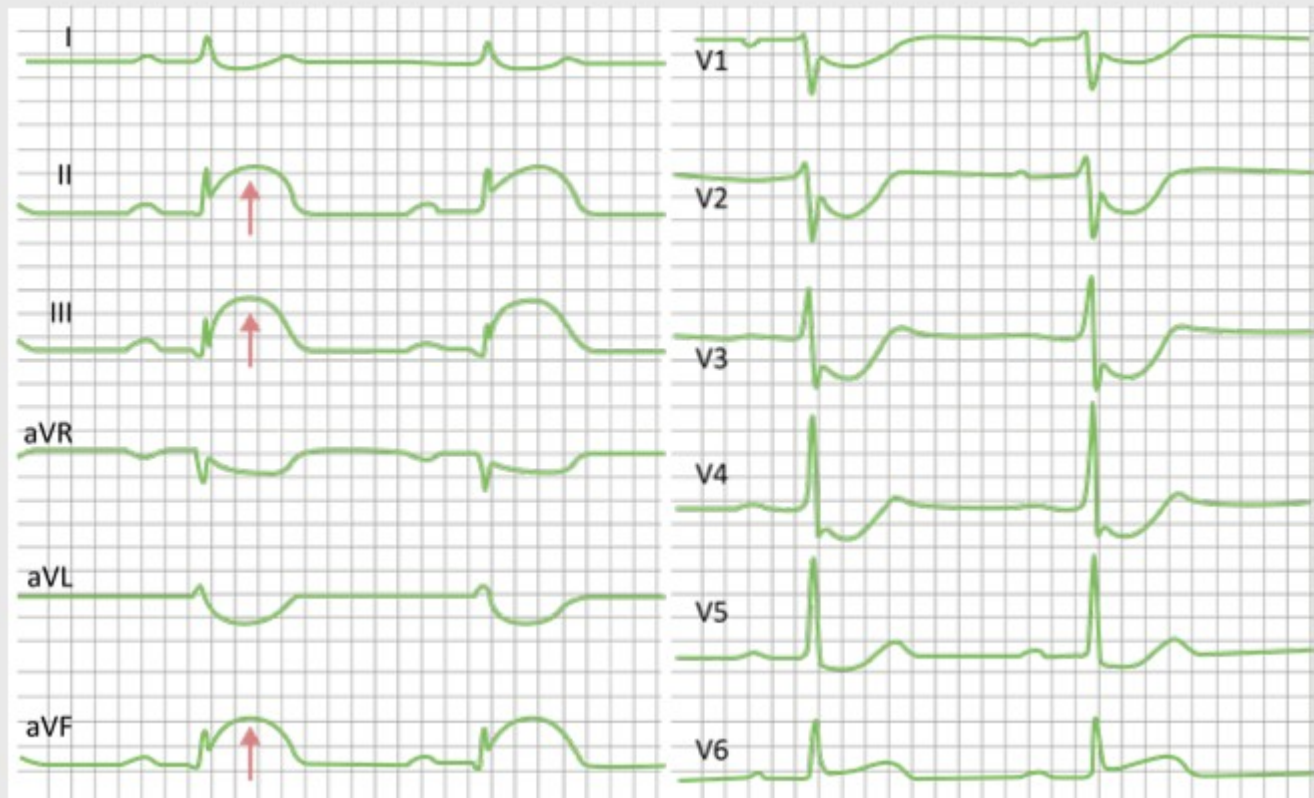
Myocardial Infarction

- This acute event is usually accompanied by an increase in cardiac enzymes, typical ECG changes, and pain symptoms, or a thrombus or wall motion abnormality that is detected by means of medical imaging.

- 
- **Non-ST-Segment Elevation Myocardial Infarction (NSTEMI)**
 - ✓ Typical symptoms of myocardial infarction persisting > 20 minutes
 - ✓ Normal, or non-specific ECG changes: ST depression and T wave inversion
 - ✓ Mildly increased cardiac enzymes
 - **ST-Segment Elevation Myocardial Infarction (STEMI)**
 - ✓ Typical symptoms of myocardial infarction persisting > 20 minutes
 - ✓ ST-segment elevation, or new-onset LBBB
 - ✓ Markedly increased cardiac enzymes

- 
- The typical symptoms include severe retrosternal pain that radiates to the left or right arm, neck, and upper jaw and does not improve after nitroglycerin administration.

ECG diagnostics for myocardial infarction



VALVULAR HEART DISEASES

- Diseases of the heart valves include a diverse group of acquired and congenital lesions
- Valvular disease results in **stenosis or insufficiency (regurgitation or incompetence), or both.**
- Stenosis is the failure of a valve to open completely, obstructing forward flow.
- Valvular stenosis is almost always due to a primary cuspal abnormality stemming from a chronic process (e.g., calcification or valve scarring).
- Abnormal valves are more susceptible to infection and thus predispose patients to infective endocarditis.
- Valvular abnormalities can be congenital or acquired.
- By far the most common congenital valvular lesion is a bicuspid aortic valve, containing only two functional cusps instead of the normal three

Mitral Valve Disease

Mitral Stenosis

Postinflammatory scarring
(rheumatic heart disease)

Mitral Regurgitation

Abnormalities of leaflets and
commissures
Postinflammatory scarring
Infective endocarditis
Mitral valve prolapse
“Fen-phen”—induced valvular
fibrosis
Abnormalities of tensor apparatus
Rupture of papillary muscle
Papillary muscle dysfunction
(fibrosis)
Rupture of chordae tendineae
Abnormalities of left ventricular
cavity and/or annulus
Left ventricular enlargement
(myocarditis, dilated
cardiomyopathy)
Calcification of mitral ring

Aortic Valve Disease

Aortic Stenosis

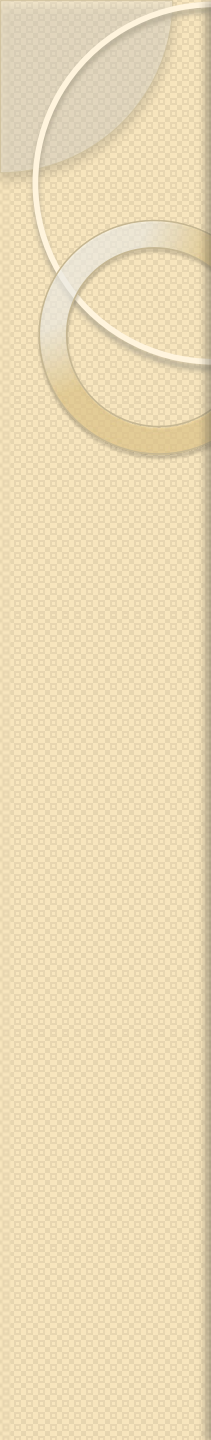
Postinflammatory scarring
(rheumatic heart disease)
Senile calcific aortic stenosis
Calcification of congenitally
deformed valve

Aortic Regurgitation

Intrinsic valvular disease
Postinflammatory scarring
(rheumatic heart
disease)
Infective endocarditis
Aortic disease
Degenerative aortic
dilation
Syphilitic aortitis
Ankylosing spondylitis
Rheumatoid arthritis
Marfan syndrome

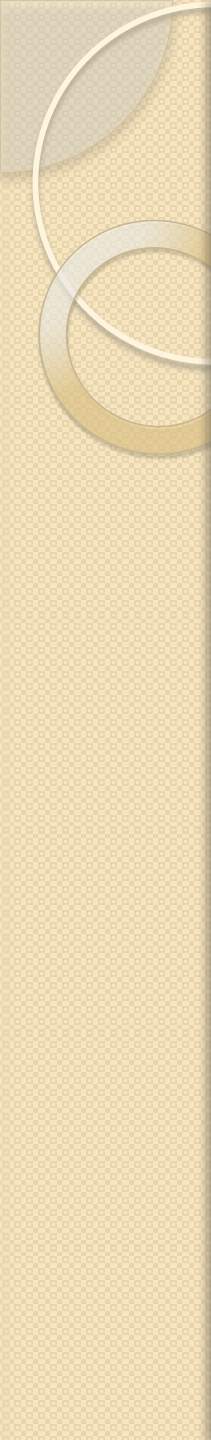
Calcific Aortic Stenosis

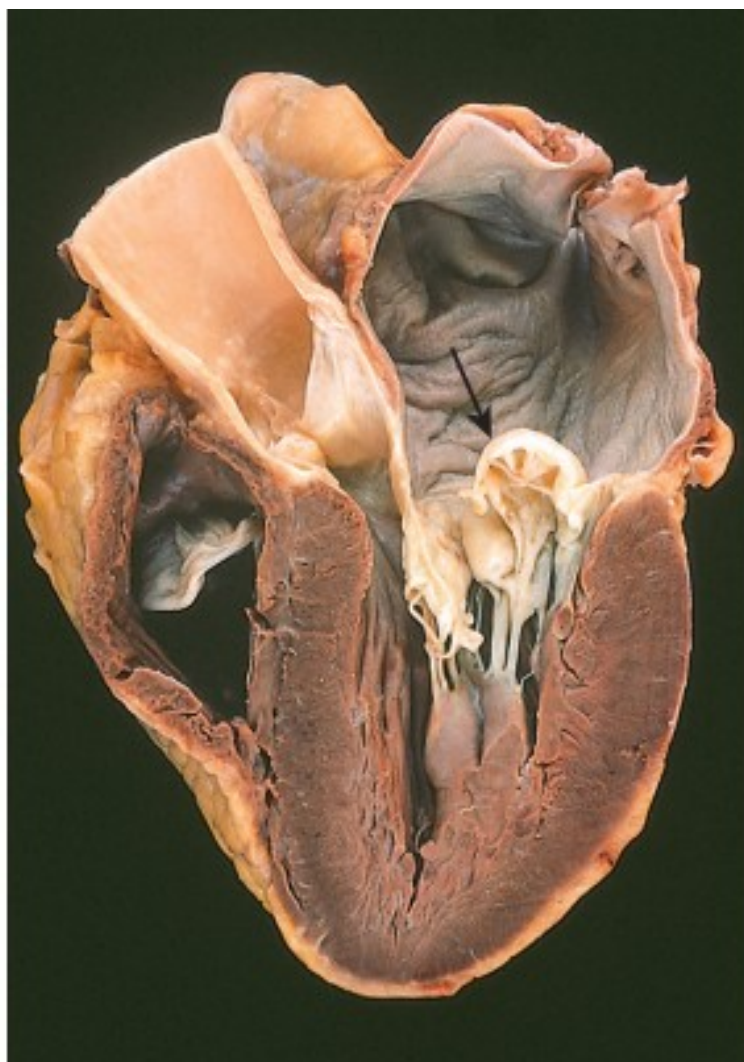
- Calcific aortic degeneration is the most common cause of aortic stenosis.
- In most cases, calcific degeneration is asymptomatic and is discovered only incidentally
- In anatomically normal valves, it typically begins to manifest when patients reach their **70s and 80s**; onset with bicuspid aortic valves is at a much earlier age (often **40 to 50 years of age**)
- The hallmark of calcific aortic stenosis is **heaped-up calcified masses** on the outflow side of the cusps; these protrude into the sinuses of Valsalva and mechanically impede valve opening

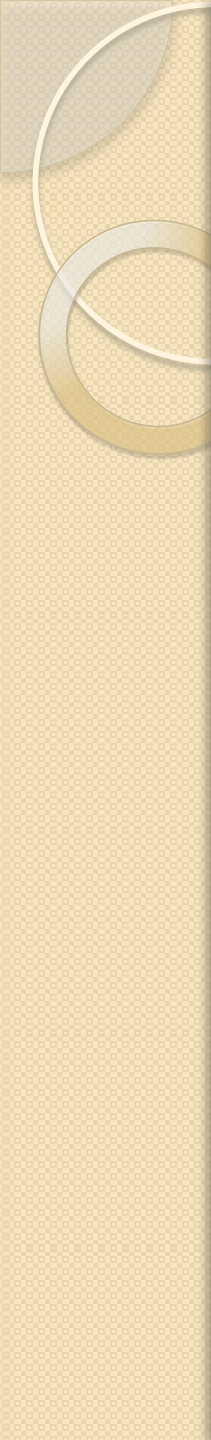
- 
- Cardiac output is maintained only by virtue of **concentric left ventricular hypertrophy**
 - The hypertrophied myocardium is prone to ischemia, and angina may develop.
 - Systolic and diastolic dysfunction collude to cause CHF, and cardiac decompensation eventually ensues.
 - The development of **angina, CHF, or syncope** in aortic stenosis heralds the exhaustion of compensatory cardiac hyperfunction and carries a poor prognosis; without surgical intervention, **50% to 80% of patients die within 2 to 3 years.**

Myxomatous Mitral Valve

- In myxomatous degeneration of the mitral valve, one or both mitral leaflets are **“floppy” and prolapse**—they balloon back into the left atrium during systole.
- Primary mitral valve prolapse is a form of myxomatous mitral degeneration affecting some 0.5% to 2.4% of adults; thus, it is one of the most common forms of valvular heart disease, with women affected almost 7-fold more often than men.

- 
- The basis for primary myxomatous degeneration of the mitral valve is unknown.
 - Nevertheless, an underlying (possibly systemic) intrinsic defect of connective tissue synthesis or remodeling is likely.
 - Thus, myxomatous degeneration of the mitral valve is a common feature of Marfan syndrome (due to fibrillin-I mutations)
 - On histologic examination, the essential change is thinning of the valve layer known as the fibrosa layer of the valve, on which the structural integrity of the leaflet depends, accompanied by expansion of the middle spongiosa layer owing to increased deposition of myxomatous (mucoid) material.



- 
- Most patients are asymptomatic, and the valvular abnormality is discovered incidentally.
 - In a minority of cases, patients complain of **palpitations, dyspnea, or atypical chest pain.**
 - Auscultation discloses a **mid systolic click**, caused by abrupt tension on the redundant valve leaflets and chordae tendineae as the valve attempts to close; there is sometimes an associated **regurgitant murmur.**
 - Approximately 3% of patients develop complications such as hemodynamically significant **mitral regurgitation and CHF**, particularly if the chordae or valve leaflets rupture.
 - Patients with primary myxomatous degeneration also are at increased risk for the development of infective endocarditis (see later), as well as SCD due to ventricular arrhythmias.
 - Stroke or other systemic infarctions may rarely occur from embolism of thrombi formed in the left atrium.

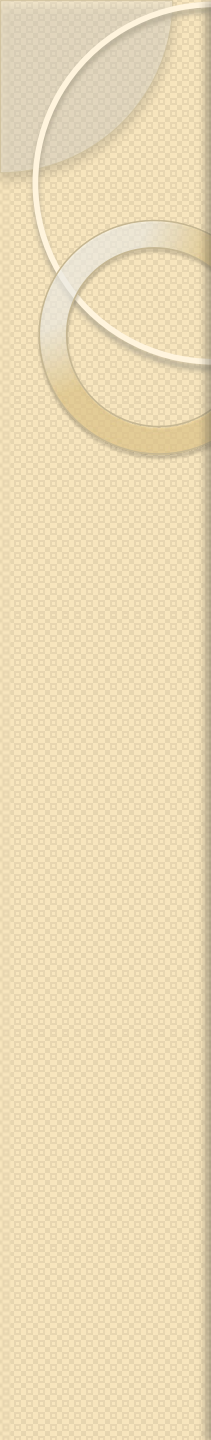
Rheumatic fever and chronic rheumatic heart disease

Rheumatic Fever

- Rheumatic fever is an acute, immunologically mediated, multisystem inflammatory disease that occurs after group A β -hemolytic streptococcal infections (usually pharyngitis, but also occasionally infections at other sites, such as skin).
- Rheumatic heart disease is the cardiac manifestation of rheumatic fever.
- It is associated with inflammation of all parts of the heart, but valvular inflammation and scarring produce the most important clinical features.

ETIOLOGY.

- evidence to support the link between GAS pharyngitis and acute rheumatic fever and CRHD.
- As many as $\frac{2}{3}$ of the patients have a history of URTI
 - ↪ the peak age and seasonal incidence of acute rheumatic fever closely parallel those of GAS infections.
 - ↪ Patients almost always have serologic evidence of a recent GAS infection.

- 
- The incidence of rheumatic fever (and thus rheumatic heart disease) has declined remarkably in many parts of the Western world over the past several decades due to a combination of **improved socioeconomic conditions, rapid diagnosis and treatment** of streptococcal pharyngitis, and a fortuitous (and unexplained) **decline in the virulence** of many strains of group A streptococci

Rheumatic fever...

- **Not all of the serotypes of GAS can cause rheumatic fever.**
- - **certain serotypes of GAS (M types 1, 3, 5, 6, 18, 24) are more frequently isolated than are other serotypes.**

EPIDEMIOLOGY.

- **In some developing countries, the annual incidence is currently as high as 282/100,000 population.**
- **Worldwide, CRHD remains the most common form of acquired heart disease in all age groups.**
- **as much as 50% of all cardiovascular disease and as much as 50% of all cardiac admissions in many developing countries.**

Rheumatic fever...

Development of acute rheumatic fever is dependent on various host factors.

- specific characteristics of the infecting GAS
- 5–15 yr of age, the age of greatest risk for GAS pharyngitis.
- Previous attack of acute rheumatic fever-recurrence rate of 30%
- ? genetic predisposition to acute rheumatic fever- a higher concordance rate in monozygotic twins.
- ? an association between the presence of both specific HLA markers and a specific B-cell alloantigen (D8/I7) and susceptibility to acute rheumatic fever.

Rheumatic fever...

▶ **PATHOGENESIS.**

- ▶ The pathogenic link between a GAS URTI and an attack of acute rheumatic fever is still not clear.
- ▶ Obstacle-the inability to establish an animal model. **Several theories**, two are most accepted.

I. **An immune-mediated pathogenesis**

- the immunologic cross reactivity -Common antigenic determinants are shared between certain components of GAS (M protein, protoplast membrane, cell wall group A carbohydrate, capsular hyaluronate) and specific mammalian tissues (e.g., heart, brain, joint).
- clinical similarity to other immune-mediated illnesses
- latency period b/n infection and acute rheumatic fever

Rheumatic fever...

2. The cytotoxicity theory

- ↪ suggests that a GAS toxin may be involved in the pathogenesis of acute rheumatic fever.
- ↪ GAS produces several enzymes that are cytotoxic for mammalian cardiac cells, such as streptolysin O, which has a direct cytotoxic effect on mammalian cells in tissue culture.
- ↪ can not explain the latency period between pharyngitis and acute rheumatic fever.

Rheumatic fever...

- **CLINICAL MANIFESTATIONS AND DIAGNOSIS.**

No clinical or laboratory finding is pathognomonic

- T. Duckett Jones in 1944 proposed guidelines to aid in diagnosis and to limit over diagnosis.
- The Jones criteria are intended only for the diagnosis of the initial attack of acute rheumatic fever.
- ↪ 5 major and 4 minor criteria and an absolute requirement for evidence (microbiologic or serologic) of recent GAS infection.
- ↪ 2 major criteria OR 1 major + 2 minor criteria and meets the absolute requirement- for diagnosing acute rheumatic fever.

Rheumatic fever...

- ▶ **There are 3 circumstances in which the diagnosis of acute rheumatic fever can be made without strict adherence to the Jones criteria.**
 1. Chorea may occur as the only manifestation of acute rheumatic fever.
 2. indolent carditis may be the only manifestation in patients who 1st come to medical attention months after the onset of acute rheumatic fever.
 3. recurrences of acute rheumatic fever

TABLE 182-2 -- Guidelines for the Diagnosis of Initial Attack of Rheumatic Fever (Jones Criteria, Updated 1992)

MAJOR MANIFESTATIONS ^[*]	MINOR MANIFESTATIONS	SUPPORTING EVIDENCE OF ANTECEDENT GROUP A STREPTOCOCCAL INFECTION
Carditis	Clinical features:	Positive throat culture or rapid streptococcal antigen test
Polyarthrititis	Arthralgia	
	Fever	Elevated or increasing streptococcal antibody titer
Erythema marginatum	Laboratory features:	
Subcutaneous nodules	Elevated acute phase reactants:	
	Erythrocyte sedimentation rate	
	C-reactive protein	
	Prolonged PR interval	
Chorea		

From Jones criteria, updated 1992. JAMA 1992;268:2069–2073. Copyright American Medical Association.

* The presence of 2 major or of 1 major and 2 minor manifestations indicates a high probability of acute rheumatic fever if supported by evidence of preceding group A streptococcal infection.

Rheumatic fever...

▶ Major Manifestations:

▶ I. **CARDITIS.**

- Carditis and resultant CRHD are the most serious manifestations of acute rheumatic fever .
- Rheumatic carditis is characterized by pancarditis, with active inflammation of myocardium, pericardium, and endocardium.
- Cardiac involvement during acute rheumatic fever varies in severity from fulminant, potentially fatal exudative pancarditis to mild, transient cardiac involvement.
- Endocarditis (valvulitis), which manifests by 1 or more cardiac murmurs, is a universal finding in rheumatic carditis, whereas the presence of pericarditis or myocarditis is variable.

Rheumatic fever...

- ▶ Myocarditis and/or pericarditis without evidence of endocarditis is rarely due to rheumatic heart disease.
- ▶ Most cases consist of either isolated mitral valvular disease or combined aortic and mitral valvular disease.
- ▶ Isolated aortic or right-sided valvular involvement is uncommon.
- ▶ Serious and long-term illness is related to **valvular heart disease** as a consequence of a single attack or recurrent attacks of acute rheumatic fever.
- ▶ Valvular insufficiency is characteristic of both acute and convalescent stages of acute rheumatic fever, whereas valvular stenosis usually appears several years or even decades after the acute illness.

Rheumatic fever...

- ▶ In developing countries , where acute rheumatic fever often occurs at a younger age, MS and AS may develop sooner and can occur in young children.

Acute rheumatic carditis usually presents as

- tachycardia ,cardiac murmurs,cardiomegaly ,CHF with hepatomegaly and peripheral and pulmonary edema.
- Echocardiographic findings include pericardial effusion, decreased ventricular contractility, and aortic and/or mitral regurgitation.
- Echocardiographic demonstration of regurgitant lesion without accompanying auscultatory evidence does not satisfy the Jones criteria for carditis.

Rheumatic fever...

- Carditis occurs in about 50–60% of all cases of acute rheumatic fever.
- Recurrent attacks of acute rheumatic fever in patients who had carditis with the initial attack are associated with high rates of carditis(30%)
- The major consequence of acute rheumatic carditis is chronic, progressive valvular disease, particularly valvular stenosis, which require valve replacement and predispose to infective endocarditis.

Rheumatic fever...

MIGRATORY POLYARTHRITIS.

- ▶ Arthritis occurs in about 75% of patients.
- ▶ involves larger joints, particularly the knees, ankles, wrists, and elbows.
- ▶ Involvement of the spine, small joints of the hands and feet, or hips is uncommon.
- ▶ Rheumatic joints are generally hot, red, swollen, and exquisitely tender.
- ▶ The pain can precede and can appear to be disproportionate to the other findings.
- ▶ The joint involvement is characteristically migratory in nature; a severely inflamed joint can become normal within 1–3 days without treatment, as 1 or more other large joints become involved.

Rheumatic fever...

- ▶ If you suspect acute rheumatic fever with arthritis **-withhold salicylates and observe for migratory progression.**
- ▶ A dramatic response to even small doses of salicylates-absence of such a response should suggest an alternative diagnosis.
- ▶ Rheumatic arthritis is typically **not deforming.**
- ▶ **Synovial fluid** in acute rheumatic fever usually has 10,000–100,000 white blood cells/mm³ with a predominance of neutrophils, a protein of about 4 g/dL, a normal glucose, and forms a good mucin clot.
- ▶ arthritis is the earliest manifestation of acute rheumatic fever.
- ▶ There is inverse relationship between the severity of arthritis and the severity of cardiac involvement.

Rheumatic fever...

- **CHOREA.**

- Sydenham chorea occurs in about 10–15% of cases and usually presents as an isolated, frequently subtle, neurologic behavior disorder.
- Hypotonia, Emotional lability, and incoordination are the triads of Sydenham's chorea.
- poor school performance, uncontrollable movements, and facial grimacing, exacerbated by stress and disappearing with sleep, are characteristic.
- Chorea occasionally is unilateral.
- The latent period from acute GAS infection to chorea is usually longer.

Rheumatic fever...

- ▶ Onset can be insidious, with symptoms being present for several months before recognition.

Clinical maneuvers to elicit features of chorea

1. demonstration of milkmaid's grip (irregular contractions of the muscles of the hands while squeezing the examiner's fingers)
2. spooning and pronation of the hands when the patient's arms are extended
3. wormian darting movements of the tongue upon protrusion
4. examination of handwriting to evaluate fine motor movements.

chorea rarely leads to permanent neurologic sequelae

Rheumatic fever...

ERYTHEMA MARGINATUM.

- ▶ Erythema marginatum is a rare (<3% of patients with acute rheumatic fever) but characteristic rash of acute rheumatic fever.
- ▶ It consists of erythematous, serpiginous, macular lesions with pale centers that are not pruritic.
- ▶ It occurs primarily on the trunk and extremities, but not on the face, and it can be accentuated by warming the skin.

SUBCUTANEOUS NODULES.

- ▶ Subcutaneous nodules are a rare ($\leq 1\%$ of patients with acute rheumatic fever) and consist of firm nodules approximately 1 cm in diameter along the extensor surfaces of tendons near bony prominences.
- ▶ There is a correlation between the presence of these nodules and significant rheumatic heart disease.

Rheumatic fever...

▶ **Minor Manifestations.**

=The 2 clinical minor manifestations are

- arthralgia (in the absence of polyarthrititis as a major criterion) and
- fever (typically temperature $\geq 102^{\circ}\text{F}$ and occurring early in the course of illness).

=The 2 laboratory minor manifestations are

- elevated acute-phase reactants (e.g., CRP, ESR)
- prolonged PR interval on ECG.

*However, a prolonged PR interval alone does not constitute evidence of carditis or predict long-term cardiac sequelae.

Rheumatic fever...

- **Recent Group A Streptococcus Infection-** an **absolute requirement** for the diagnosis of acute rheumatic fever is supporting evidence of a recent GAS infection.
 - ▢ develops 2–4 wk after an acute episode of GAS pharyngitis at a time CI fxs of pharyngitis disappear and only 10–20% of the throat culture or rapid streptococcal antigen test results are positive.
 - ▢ One third of patients have no history of an antecedent pharyngitis.
 - ▢ Therefore, evidence of an antecedent GAS infection is usually based on elevated or increasing serum antistreptococcal antibody titers.

Rheumatic fever...

-A slide agglutination (Streptozyme) test for detection of GAS antigens

↪ purported to detect antibodies against 5 different GAS antigens.

↪ If only a single antibody is measured (usually ASO), only 80–85% of patients with acute rheumatic fever have an elevated titer

↪ however, 95–100% have an elevation if 3 different antibodies (antistreptolysin O, anti-DNase B, antihyaluronidase) are measured.

↪ Except for patients with chorea, clinical findings of acute rheumatic fever generally coincide with peak antistreptococcal antibody responses.

Rheumatic fever...

► **TREATMENT.**

1. Supportive

- bed rest and monitored closely for evidence of carditis.
- They can be allowed to ambulate as soon as the signs of acute inflammation have subsided.
- patients with carditis require longer periods of bed rest.

2. Antibiotic Therapy.

- should receive 10 days of orally administered penicillin or erythromycin, or a single intramuscular injection of benzathine penicillin to eradicate GAS from the upper respiratory tract.

Regardless of throat swab result

- After this initial course of antibiotic therapy, the patient should be started on long-term antibiotic prophylaxis.

Rheumatic fever...

3. Anti-Inflammatory Therapy:

- Agents such as acetaminophen can be used to control pain and fever while the patient is being observed for more definite signs of acute rheumatic fever.

- a. oral salicylates-** Patients with typical migratory polyarthrititis and those with carditis without cardiomegaly or congestive heart failure

- ↪ The usual dose of aspirin is 100 mg/kg/day in 4 divided doses PO for 3–5 days, followed by 75 mg/kg/day in 4 divided doses PO for 4 wk.

- b. corticosteroids** -Patients with carditis and cardiomegaly or congestive heart failure.

Rheumatic fever...

- ▶ The usual dose of prednisone is 2 mg/kg/day in 4 divided doses for 2–3 wk followed by a tapering of the dose that reduces the dose by 5 mg/24 hr every 2–3 days.
- ▶ At the beginning of the tapering, aspirin should be started at 75 mg/kg/day in 4 divided doses for 6 wk.

3. RX of heart failure. digoxin, fluid and salt restriction, diuretics, and oxygen- for treatment of CHF in moderate to severe carditis. Be careful about digoxin toxicity.

Termination of the anti-inflammatory therapy may be followed by rebounds phenomena which should be left untreated **unless** the clinical manifestations are **severe**; salicylates or steroids should be reinstated in such cases.

Rheumatic fever...

Rx of Sydenham Chorea.

- anti-inflammatory agents are usually not indicated.
- Sedatives may be helpful early in the course of chorea; phenobarbital (16–32 mg every 6–8 hr PO)
- If phenobarbital is ineffective, then haloperidol (0.01–0.03 mg/kg/24 hr divided bid PO) or chlorpromazine (0.5 mg/kg every 4–6 hr PO) should be initiated.

COMPLICATIONS.

- ▶ The arthritis and chorea of acute rheumatic fever resolve completely without sequelae.
- ▶ The long-term sequelae of rheumatic fever are usually limited to the heart
- ▶ Patients with VHD secondary to acute rheumatic fever are at increased risk for developing infective endocarditis during episodes of transient bacteremia.

Rheumatic fever...

▶ **PROGNOSIS.**

- ▶ depends on the clinical manifestations present at the initial episode, the severity of the initial episode, and the presence of recurrences.
- ▶ Approx 70% of the patients with carditis- recover with no residual heart disease; the more severe the initial cardiac involvement, the greater the risk for residual heart disease.
- ▶ Patients without carditis during the initial episode are unlikely to have carditis with recurrences.
- ▶ patients with carditis during the initial episode are likely to have carditis with recurrences, and the risk for permanent heart damage increases with each recurrence.

Rheumatic fever...

- ▶ Patients who have had acute rheumatic fever are susceptible to recurrent attacks following reinfection of the upper respiratory tract with GAS.
- ▶ **Without antibiotic prophylaxis, 75% of patients** who had an initial episode of acute rheumatic fever **had 1 or more recurrences during their lifetime**-recurrences are a major source of morbidity and mortality.
- ▶ The risk of recurrence is highest immediately after the initial episode and decreases with time.
- ▶ 20% of patients who present with “pure” chorea who are not given secondary prophylaxis develop rheumatic heart disease within 20 yr.
- ▶ Therefore, require long-term antibiotic prophylaxis.

Rheumatic fever...

▶ **PREVENTION.**

1. Primary Prevention.

- ▶ Appropriate antibiotic therapy instituted before the 9th day of symptoms of acute GAS pharyngitis is highly effective in preventing 1st attacks of acute rheumatic fever from that episode.

2. Secondary Prevention- to prevent recurrence

- ▶ Secondary prevention is directed at preventing acute GAS pharyngitis in patients at substantial risk of recurrent acute rheumatic fever.
- ▶ Secondary prevention requires continuous antibiotic prophylaxis every 4 wks.

3. Tertiary prevention- involves prevention of further cardiac deterioration in patients with established CRHD

Chronic rheumatic heart disease

- ▶ Rheumatic involvement of the valves and endocardium is the most important manifestation of rheumatic fever.
- ▶ The valvular lesions begin as small verrucae composed of fibrin and blood cells along the borders of one or more of the heart valves.
- ▶ The mitral valve is affected most often, followed in frequency by the aortic valve; right-sided heart manifestations are rare.
- ▶ As the inflammation subsides, the verrucae tend to disappear and leave scar tissue.
- ▶ With repeated attacks of rheumatic fever, new verrucae form near the previous ones, and the mural endocardium and chordae tendineae become involved

Chronic rheumatic heart...

- ▶ PATTERNS OF VALVULAR DISEASE

- ▶ **MITRAL INSUFFICIENCY**

- ▶ Pathophysiology.

- ▶ **Mitral insufficiency is the result of structural changes that usually include some loss of valvular substance and shortening and thickening of the chordae tendineae.**
- ▶ **During acute rheumatic fever with severe cardiac involvement, heart failure is caused by a combination of mitral insufficiency coupled with inflammatory disease of the pericardium, myocardium, endocardium, and epicardium.**
- ▶ **Because of the high volume load and inflammatory process, the left ventricle becomes enlarged.**

Chronic rheumatic heart...

- ▶ The left atrium dilates as blood regurgitates. Increased left atrial pressure results in pulmonary congestion and symptoms of left-sided heart failure.
- ▶ Spontaneous improvement usually occurs with time, even in patients in whom mitral insufficiency is severe at the onset.
- ▶ The resultant chronic lesion is most often mild or moderate in severity, and the patient is asymptomatic.
- ▶ More than half of patients with acute mitral insufficiency no longer have the mitral murmur 1 yr later.
- ▶ In patients with severe chronic mitral insufficiency, pulmonary arterial pressure becomes elevated, the right ventricle and atrium become enlarged, and right-sided heart failure subsequently develops.

Chronic rheumatic heart...

- ▶ Clinical Manifestations.

- ▶ **The physical signs of mitral insufficiency depend on its severity.**

With mild disease, signs of heart failure are not present auscultation reveals a high-pitched holosystolic murmur at the apex that radiates to the axilla.

With severe mitral insufficiency-

- Cardiomegally with a heaving apical left ventricular impulse and often an apical systolic thrill.**

- S2 is accentuated, A 3rd heart sound is prominent.**

- ↪ **A short mid-diastolic rumbling murmur is caused by increased blood flow across the mitral valve as a result of the insufficiency (not necessarily MS)**

Chronic rheumatic heart...

▶ Lab

ECG -shows prominent bifid P waves

- signs of left ventricular hypertrophy
- associated right ventricular hypertrophy if pulmonary hypertension is present.

CXR- prominence of the left atrium and ventricle(straightening of left heart border and widening of the carinal angle >90 degree)

- Congestion of perihilar vessels (if pulmonary hypertension)

Echocardiography -shows enlargement of the left atrium and ventricle, and Doppler studies demonstrate the severity of the mitral regurgitation.

Chronic rheumatic heart...

- ▶ Complications.
 - ↖ **cardiac failure ,atrial fibrillation, or infective endocarditis.**
 - ↖ **The effects of chronic mitral insufficiency may become manifest after many years and include right ventricular failure and atrial and ventricular arrhythmias.**
- ▶ Treatment.
- ▶ **In patients with mild mitral insufficiency, prophylaxis against recurrences of rheumatic fever is all that is required.**
- ▶ **Afterload -reducing agents (ACE inhibitors) may reduce the regurgitant volume and preserve left ventricular function.**
- ▶ **Surgical treatment**

Chronic rheumatic heart...

▶ **MITRAL STENOSIS**

- ▶ Pathophysiology.
- ▶ Mitral stenosis of rheumatic origin results from fibrosis of the mitral ring, commissural adhesions, and contracture of the valve leaflets, chordae, and papillary muscles over time.
- ▶ It takes 10 yr or more for the lesion to become fully established, although the process may occasionally be accelerated.
- ▶ Significant MS results in increased pressure and enlargement and hypertrophy of the left atrium, pulmonary venous hypertension, increased pulmonary vascular resistance, and pulmonary hypertension.
- ▶ Right ventricular and atrial dilatation and hypertrophy ensue and are followed by right-sided heart failure.

Chronic rheumatic heart...

- ▶ Clinical Manifestations.
- ▶ Patients with mild lesions are asymptomatic.
- ▶ More severe degrees of obstruction are associated with exercise intolerance and dyspnea.
- ▶ Critical lesions can result in orthopnea, paroxysmal nocturnal dyspnea, and overt pulmonary edema, as well as atrial arrhythmias.
- ▶ pulmonary hypertension -right ventricular dilatation may result in functional tricuspid insufficiency, hepatomegaly, ascites, and edema.
- ▶ Hemoptysis caused by rupture of bronchial or pleurohilar veins and, occasionally, by pulmonary infarction may occur.
- ▶ Jugular venous pressure may be increased

Chronic rheumatic heart...

- parasternal right ventricular lift is palpable when pulmonary pressure is high.
- a loud 1st heart sound, an opening snap of the mitral valve, and a long, low-pitched, rumbling mitral diastolic murmur with presystolic accentuation at the apex.
- A holosystolic murmur secondary to tricuspid insufficiency may be audible.
- If pulmonary hypertension, P2 is accentuated.
- An early diastolic murmur may be caused by secondary pulmonary valvular insufficiency.

Chronic rheumatic heart...

▶ **AORTIC INSUFFICIENCY.**

- ▶ Regurgitation of blood leads to volume overload with dilatation and hypertrophy of the left ventricle. Combined mitral and aortic insufficiency is more common than aortic involvement alone.
- ▶ **Clinical Manifestations.**
- ▶ **In severe aortic insufficiency-**
- ▶ Excessive sweating and heat intolerance .
- ▶ Dyspnea on exertion , orthopnea and pulmonary edema.
- ▶ wide pulse pressure ,bounding peripheral pulses, Systolic BP is elevated, and diastolic pressure is lowered, the heart is enlarged, with a left ventricular apical heave,A diastolic thrill may be present.

Chronic rheumatic heart...

- ▶ **The diastolic murmur is heard over the upper and mid LSB with radiation to the apex and the aortic area, high-pitched blowing in quality and is easily audible in full expiration with the patient leaning forward.**
- ▶ **A systolic ejection murmur due to the increased stroke volume.**
- ▶ **An apical presystolic murmur (Austin Flint murmur)**
- ▶ **Roentgenograms show enlargement of the left ventricle and aorta.**
- ▶ **ECG may reveals signs of LV hypertrophy and strain with prominent P waves.**
- ▶ **The echo- a large LV. Doppler studies demonstrate the degree of aortic runoff into the left ventricle..**

Chronic rheumatic heart...

- Prognosis and Treatment.
- **Mild and moderate lesions are well tolerated.**
- **Many adolescents with severe regurgitation are symptom free and tolerate advanced lesions into the 3rd–4th decades.**
- **Unlike mitral insufficiency, aortic insufficiency does not regress.**
- **Patients with combined lesions may have only aortic involvement 1–2 yr later.**
- **Treatment consists of** afterload reducers (ACE inhibitors) and prophylaxis against recurrence of acute rheumatic fever and the development of infective endocarditis.

Chronic rheumatic heart...

- **TRICUSPID VALVE DISEASE.**
- **Primary tricuspid involvement is rare.**
- **Tricuspid insufficiency is more common** secondary to right ventricular dilatation resulting from unrepaired left-sided lesions.
- **the signs produced by tricuspid insufficiency include** prominent pulsations of the jugular veins, systolic pulsations of the liver, and a blowing holosystolic murmur at the lower left sternal border that increases in intensity during inspiration.
- **Concomitant signs of mitral or aortic valve disease, with or without atrial fibrillation, are frequent.**

Chronic rheumatic heart...

- **PULMONARY VALVE DISEASE.**
- **Pulmonary insufficiency usually occurs on a functional basis secondary to pulmonary hypertension and is a late finding with severe mitral stenosis.**
- **The murmur (Graham Steell murmur) is similar to that of aortic insufficiency, but peripheral arterial signs (bounding pulses) are absent.**
- **The correct diagnosis is confirmed by two-dimensional echocardiography and Doppler studies**

Infective endocarditis

- **ETIOLOGY.**

- Viridans-type streptococci (α -hemolytic streptococci) and *Staphylococcus aureus* are the leading causative agents for endocarditis in pediatric patients.
- in $\approx 6\%$ of cases, blood cultures are negative for any organisms .
- No relationship exists between the infecting organism and the type of congenital defect, the duration of illness, or the age of the child.
- Staphylococcal endocarditis is more common in patients with no underlying heart disease;

- 
- viridans group streptococcal infection is more common after dental procedures;
 - group D enterococci are seen more often after lower bowel or genitourinary manipulation;
 - *pseudomonas aeruginosa* or *Serratia marcescens* is seen more frequently in intravenous drug users;
 - fungal organisms are encountered after open heart surgery.
 - Coagulase-negative staphylococci are common in the presence of an indwelling central venous catheter

Common: native valve or other cardiac lesion

V. Streptococci

S. Aureus

Group D streptococci (enterococci)-S. bovis and S. fecalis

Uncommon: native valve or other cardiac lesions

S. pneumoniae, *H. influenzae*, coag negative staph

Coxiella burnetii, *Chlamydia psittaci*, *C. trachomatis*, *C. pneumoniae*

Brucella, Bartonella, N. gonorrhoeae,

HACEK group

Campylobacter fetus, legionella

Culture negative (6%)

Prosthetic valve

Staph epidermidis

Viridans streptococci

S. Aureus

Pseudomonas aeruginosa

Serratia marcescens, legionella spp, HACEK group

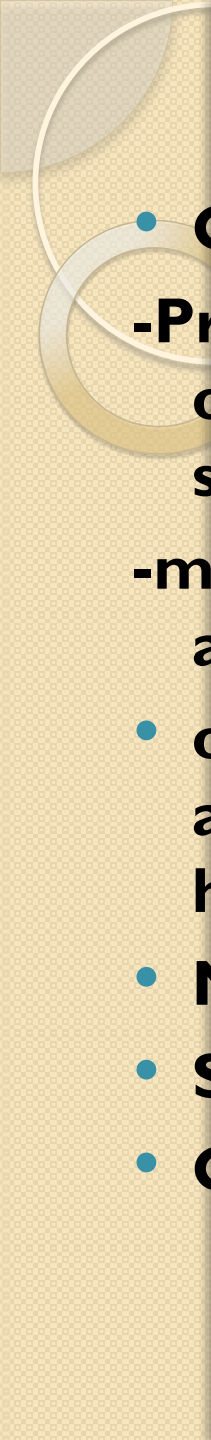
Fungi(candida, aspergillus etc)

** fungi and some fastidious bacteria may be culture -ve at first

- **EPIDEMIOLOGY.**

- **Infective endocarditis is often a complication of congenital or rheumatic heart disease but can also occur in children without any abnormal valves.**
- **Patients with lesions in which blood is ejected at high velocity through a hole or stenotic orifice are most susceptible to endocarditis.**
- **Vegetations usually form at the site of the endocardial or intimal erosion that results from the turbulent flow.**
- **In $\approx 30\%$ of patients with infective endocarditis, a predisposing factor is recognized..**

- 
- **A surgical or dental procedure can be implicated in $\approx 65\%$ of cases in which the potential source of bacteremia is identified.**
 - **Poor dental hygiene in children with cyanotic heart disease results in a greater risk for endocarditis.**
 - **Primary bacteremia with *Staphylococcus aureus* is another risk for endocarditis (10% risk).**
 - **The occurrence of endocarditis directly after heart surgery is relatively low, but it is frequently an antecedent event.**

- 
- **CLINICAL MANIFESTATIONS -2 types**
 - **Prolonged fever without other manifestations (except, occasionally, weight loss) that persists for as long as several months may be the only symptom.**
 - **may be acute and severe, with high, intermittent fever and prostration.**
 - **often nonspecific and consist of low-grade fever with afternoon elevations, fatigue, myalgia, arthralgia, headache, chills, nausea, and vomiting.**
 - **New or changing heart murmurs.**
 - **Splenomegally, petechiae**
 - **CNS manifestations- with staph aureus**

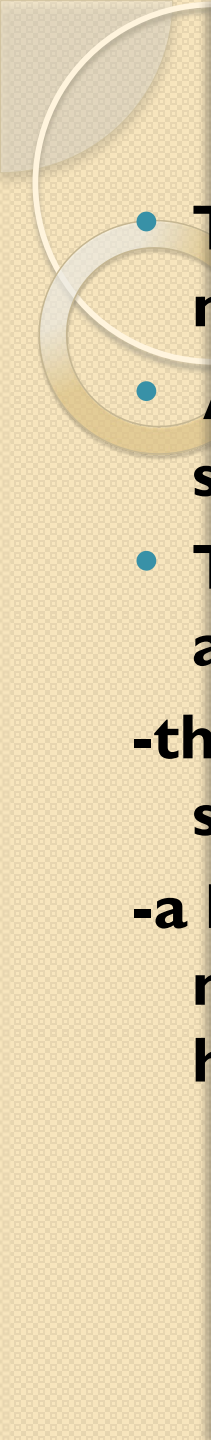
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- **Myocardial abscesses -with staphylococcal disease and may damage the cardiac conducting system, causing heart block, or may rupture into the pericardium and produce purulent pericarditis.**
 - **Pulmonary and other systemic emboli are infrequent, except with fungal disease.**
 - **Many of the classic skin findings develop late in the course of the disease are seldom seen – include Osler nodes , Janeway lesions and splinter hemorrhages -These lesions may represent vasculitis produced by circulating antigen-antibody complexes.**

- **DIAGNOSIS.**

blood culture –gold standard

- **Three to five separate blood collections should be obtained after careful preparation of the phlebotomy site.**
- **timing of collection is not impor't b/se bacteremia can be expected to be relatively constant.**
- In 90% of cases of endocarditis, the causative agent is recovered from the 1st two blood cultures.**
- pretreatment reduces the yield of culture to 50–60%.**
- Other specimens that may be cultured include scrapings from cutaneous lesions, urine, synovial fluid, abscesses, cerebrospinal fluid.**
- **Serologic diagnosis for patients with unusual bacteria.**
- **three blood culture sets (with two bottles per set), separated from each other by at least 1 h, should be obtained from different venipuncture sites over 24 h.**

- **The Duke criteria help in the diagnosis of endocarditis.**
- **Major criteria include**
- **(1) positive blood cultures (two separate cultures for a usual pathogen, two or more for less typical pathogens) and**
- **(2) echocardiography (intracardiac mass on a valve or other site, regurgitant flow near a prosthesis, abscess, partial dehiscence of prosthetic valves, or new valve regurgitant flow).**
- **Minor criteria include predisposing conditions, fever, embolic-vascular signs, immune complex phenomena (glomerulonephritis, arthritis, rheumatoid factor, Osler nodes, Roth spots), a single positive blood culture or serologic evidence of infection, and echocardiographic signs not meeting the major criteria..**

- 
- **Two major criteria, one major and three minor, or five minor criteria suggest definite endocarditis.**
 - **A modification of the Duke criteria may increase sensitivity while maintaining specificity.**
 - **The following minor criteria are added to those already listed:**
 - the presence of newly diagnosed clubbing, splenomegaly, splinter hemorrhages, and petechiae;**
 - a high ESR or CRP level; and the presence of central nonfeeding lines, peripheral lines, and microscopic hematuria**

High risk

- Prosthetic valve, previous endocarditis
- Complex cyanotic CHD(TGA, TOF, single ventricle)
- Systemic to pulmonary shunts

Moderate risk

- Other congenital heart diseases
- Rheumatic heart disease
- Hypertrophic cardiomyopathy

Negligible risk- no need for prophylaxis

- Isolated secundum ASD,
- Surgically repaired ASD, VSD or PDA(beyond 6 mo repair),
rheumatic fever without valve dysfunction,
- pacemakers

- **PROGNOSIS AND COMPLICATIONS.**
- **Despite the use of antibiotic agents, mortality remains at 20–25%. Serious morbidity occurs in 50–60% of children with documented infective endocarditis;**
 - ▢ **heart failure caused by vegetations involving the aortic or mitral valve.**
 - ▢ **Myocardial abscesses and toxic myocarditis**
 - ▢ **Systemic emboli with CNS manifestations**
 - ▢ **Pulmonary emboli in children with VSD or TOF.**
 - ▢ **mycotic aneurysms, rupture of a sinus of Valsalva, obstruction of a valve secondary to large vegetations, acquired VSD, and heart block**
 - ▢ **meningitis, osteomyelitis, arthritis, renal abscess, and immune-complex mediated glomerulonephritis**



High dose

2. Treatment of the congestive state

-use IF chart serial measurements of the following

[illegible]

4. Surgery

in severe aortic & mitral involvement with intractable heart failure, Myocardial abscess, recurrent emboli
Failure to sterilize blood and increase in vegetation size with appropriate antibiotic Rx

The antibiotics of choice for empiric treatment are

The combinations of

1. Ampicillin 300mg/kg/24hrs in 4 divided doses
2. Cloxacillin 200mg/kg/24hrs in 4 divided doses
3. Gentamycin 3-7.5mg/kg/24hrs in 3 divided doses

Gentamycin should be given only for 2wks

Infective endocarditis is unlikely if fever disappears before 4 days of treatment.

- Prevention

Antibiotic prophylaxis

1. Most patients-dental/oral/URT procedures

- oral amoxicillin 50mg/kg 1hr before procedure

If unable to take PO IV/IM ampicillin 50mg/kg 30min before

- Pen allergy oral clindamycin 20mg/kg 1hr before

2. For GU and GI procedure

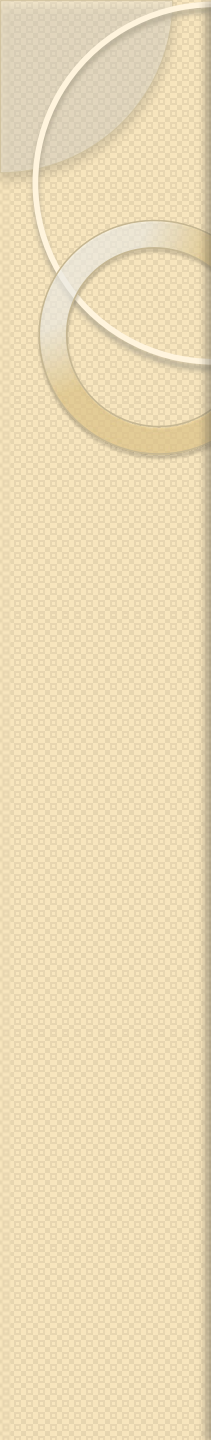
High risk- IV/IM amp 50mg/kg + genta 1.5mg/kg 30min before procedure plus 6hrs later IV/IM amp or oral amoxicillin 25mg/kg

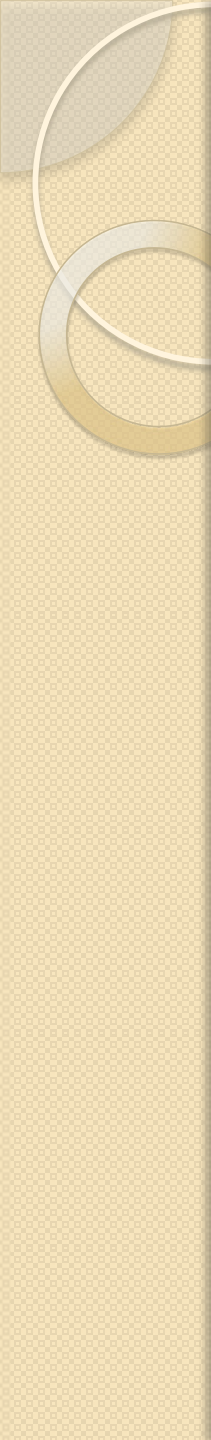
- pen allergy IV vancomycin 20mg/kg +genta(1-2hr)

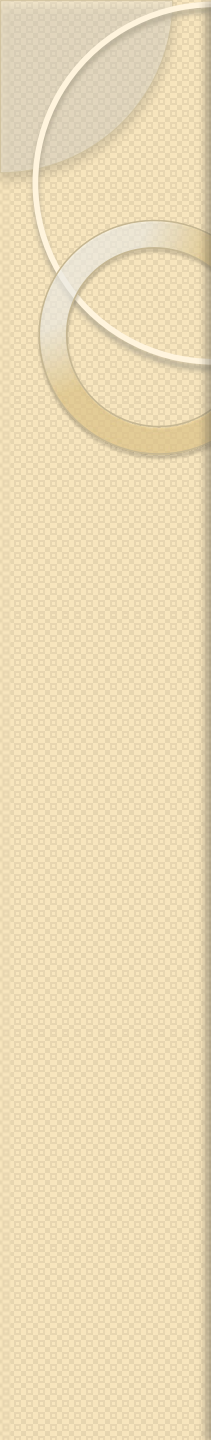
Moderate risk- oral amox 50mg/kg 1hr before or IM/IV amp 30min before; for pen allergy vancomycin 20mg/kg 1-2hrs before procedure

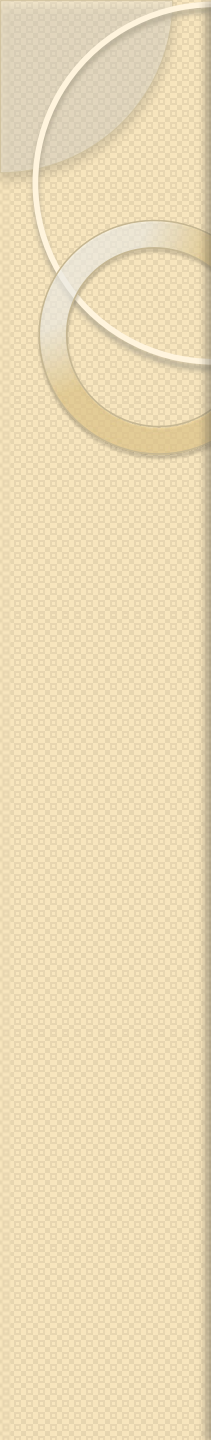
Hypertension

- Systolic blood pressure of ≥ 130 mmHg and/or diastolic blood pressure
- ≥ 90 mmHg Approximately 75 % of the world's population suffers from arterial hypertension.
- Of this about half are unaware of it.
- According to the etiology, arterial hypertension is divided into 2 types, primary hypertension (also known as essential hypertension) and secondary hypertension.

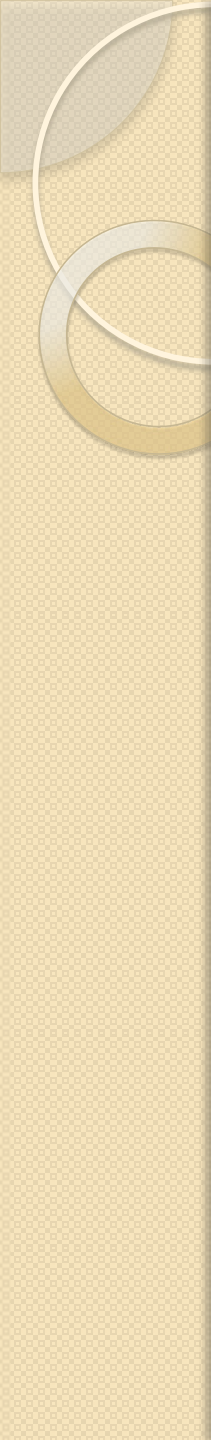
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- More than 90 % of hypertensive patients have primary hypertension.
 - This type of hypertension is idiopathic or without any known cause.
 - Diagnosis is made through the principle of exclusion.
 - There are several factors causing secondary hypertension:
 - Neurogenic, psychogenic, and iatrogenic causes; the latter includes contraceptives, NSAR drugs, consumption of illicit drugs, and toxic substances such as licorice
 - Medical conditions such as obstructive sleep apnea syndrome, coarctation of the aorta, and atherosclerosis

- 
- Another type of arterial hypertension is hypertensive disease of pregnancy.
 - Risk factors include increasing maternal age and multifetal pregnancies.
 - Hypertensive disorders of pregnancy are divided into:
 - ❑ Gestational hypertension (hypertension not associated with proteinuria)
 - ❑ Pre-eclampsia (hypertension associated with proteinuria)
 - ❑ Eclampsia (pre-eclampsia with at least 1 episode of convulsion due to cerebral dysrhythmia)

- 
- Hypertension is divided into several stages.
 - An increase in blood pressure as a result of physical exertion is not considered hypertension.
 - Accordingly, the American College of Cardiology has updated its guidelines to define hypertension in 4 different stages as follows:

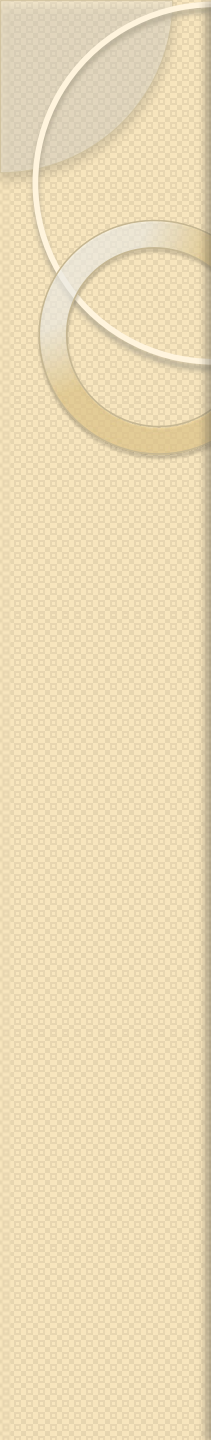


Term	Systolic BP (mmHg)	Diastolic BP (mmHg)	Notes
Normal	< 120	< 80	
Elevated	120–129	and < 80	
Stage 1	130–139	or 80–89	Treatment is needed
Stage 2	140–179	or 90 or more	Treatment is needed
Hypertensive crisis	> 180	or > 120	Patient also needs an immediate change to blood pressure control medication or should be hospitalized.

- 
- Hypertensive crises can be divided into 2 types:
 - hypertensive urgency and hypertensive emergency.
 - The former differs from the latter in that there is no organ damage in hypertensive urgency while hypertensive emergency always considers end organ damage such as encephalopathy, hemorrhagic/ischemic CVA, retinal hemorrhage/papilledema, CHF, ACS, aortic dissection, aneurysm of the abdominal aorta, ARF, hematuria, or MAHA.

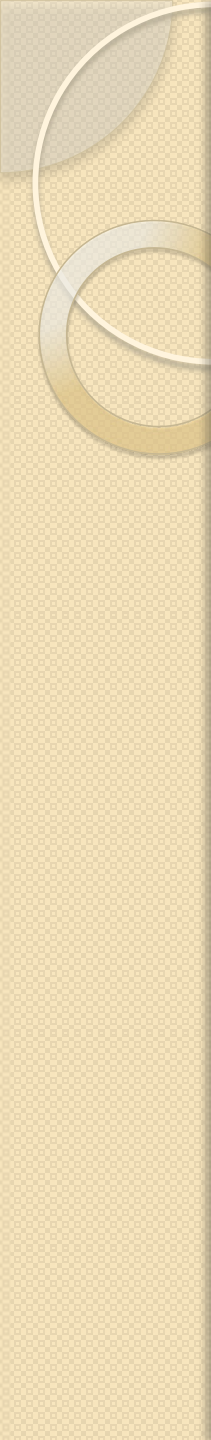
Pathophysiology of Hypertension

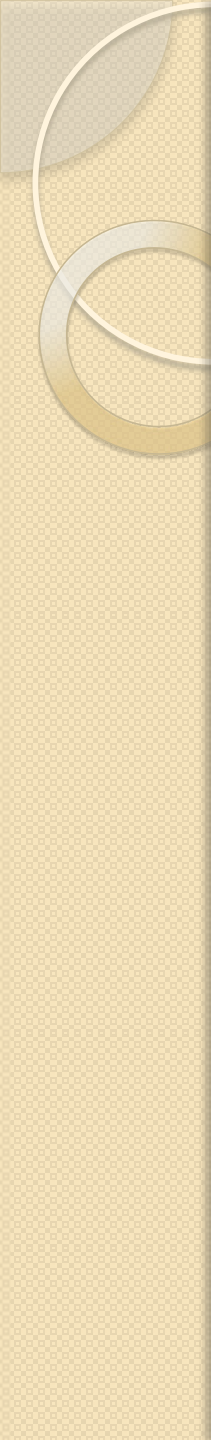
- Arterial hypertension develops due to disturbances of the regulatory mechanism that usually keeps blood pressure stable.
- Disturbances include an increased peripheral vascular resistance, increase in cardiac output, or a combination of both
- During disease progression, several compensatory mechanisms take place to consistently keep the blood pressure at an elevated level.
- In order to maintain cardiac output, the cardiac muscle hypertrophies which helps to permanently withstand the increased pressure.
- The resisting blood vessels also undergo hypertrophy.

- 
- The kidneys are involved in this compensatory process.
 - Even though renal blood flow and glomerular filtration rate are, on the whole, constant, increased sodium excretion (pressure natriuresis) accompanies the increase in blood pressure in order to counteract hypertension.

Clinical Features of Hypertension

- Symptoms of arterial hypertension frequently manifest later on.
- Typical symptoms include:
 - ☐ Early morning headaches
 - ☐ Sleep disorders, dizziness
 - ☐ Nose bleeds
 - ☐ Ringing in the ears
 - ☐ Non-specific cardiac symptoms
 - ☐ Palpitations
- Since hypertension is often asymptomatic, regular screening is necessary to prevent end-organ damage.

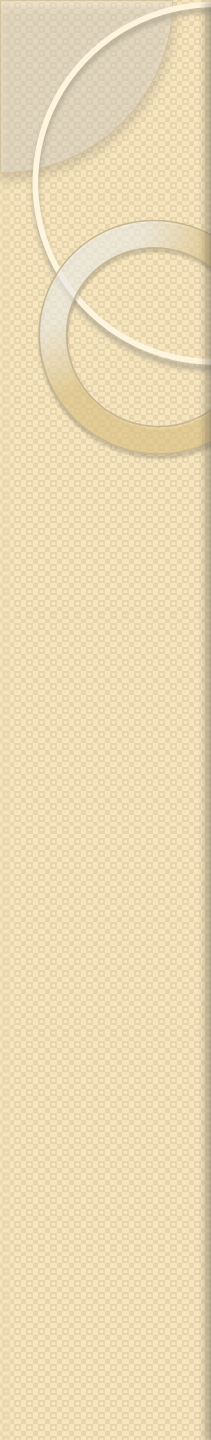
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- Subtypes and Variants of Hypertension
 - White-coat hypertension (White-coat effect):
 - Measurements of $\geq 140/90$ mmHg in the physician's office, while measurements taken at home and during blood pressure monitoring are normal
 - Diagnosed via 24-hour blood pressure monitoring

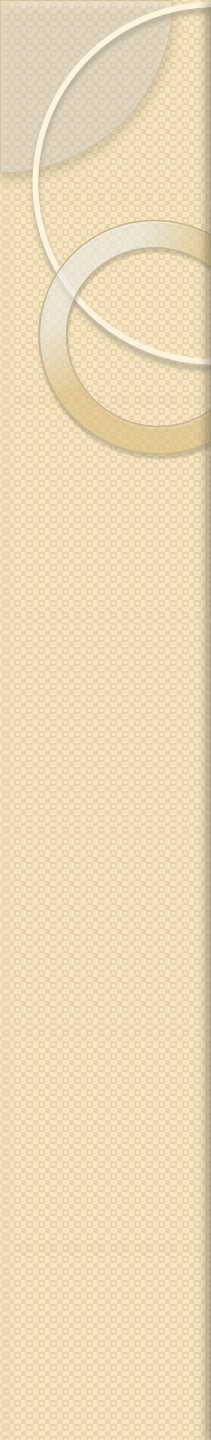
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- Isolated systolic hypertension:
 - ❑ Defined as an increase in systolic blood pressure (> 140 mmHg) with diastolic blood pressure within normal limits (< 90 mmHg)
 - ❑ Occurs in elderly populations due to decreased arterial elasticity and increased stiffness
 - ❑ Patient usually has a high risk of cardiovascular events (MI, stroke, renal dysfunction)

- Isolated ambulatory hypertension:
 - ❑ Referred to as masked hypertension; in this case, blood pressure readings at the office are normal, however, readings at home or during blood pressure monitoring are elevated to more than 140/90 mmHg.
 - ❑ This special form of hypertension may be linked to factors such as male gender and younger age, as well as smoking, alcohol consumption, and stress.
 - ❑ Where the patient is already being treated for hypertension, this condition is referred to as masked uncontrolled hypertension (MUCH).
- The focal point of diagnosing arterial hypertension is the non-invasive measurement of blood pressure according to Riva Rocci.
- Here, it is necessary to make sure that elevated blood pressure is established by taking at least 3 readings in 2 different days

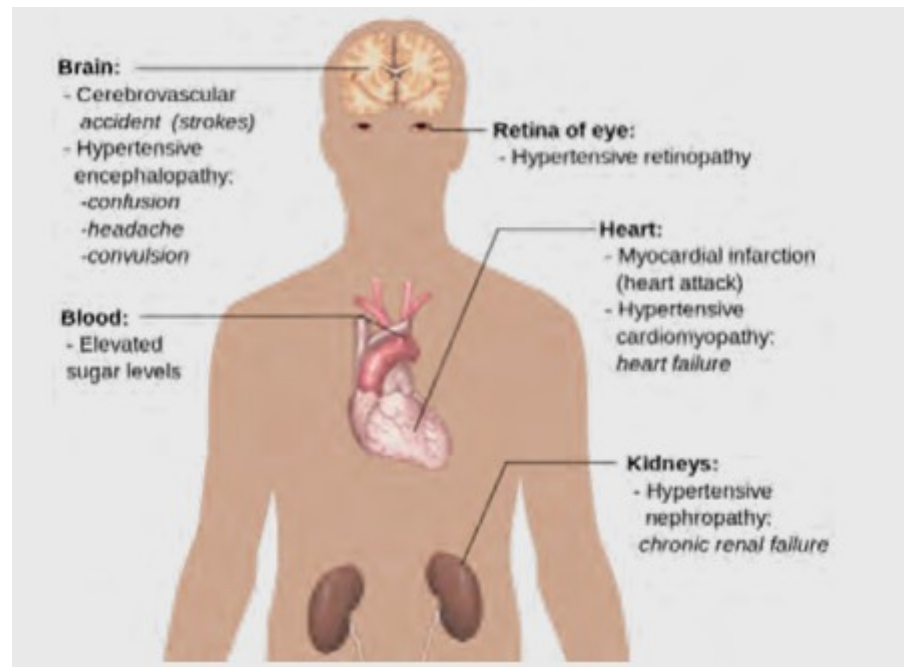
Treatment of Hypertension

- Non-pharmacological treatment (lifestyle modifications)
 - ❑ Weight reduction until a BMI of approximately 25 kg/m² has been achieved
 - ❑ Diet low in sodium, with no more than 5–6 g NaCl per day
 - ❑ Switching to a Mediterranean diet
 - ❑ Adjusting lifestyle to reduce hypertension (smoking, alcohol and coffee intake should be reconsidered)

- 
- Pharmacological treatment
 - It is recommended to start pharmacological therapy as a monotherapy.
 - In the event of blood pressure values strongly deviating from normal values ($> 130/80$ mmHg) or in cases of comorbidities, primary combination therapy should be initiated.
 - First-line medications include:
 - ☐ Thiazide diuretics
 - ☐ ACE inhibitors
 - ☐ ARBs
 - ☐ Calcium channel blockers

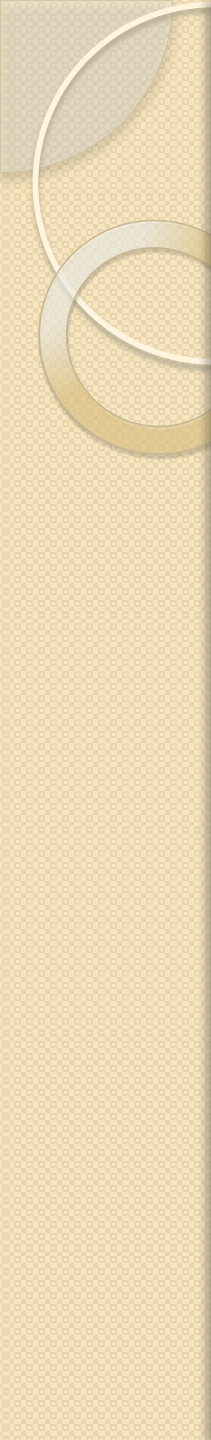
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- Secondary antihypertensives are used when primary medications do not work or if there are special indications such as in hypertensive patients with ischemic heart disease.
 - Secondary antihypertensives include:
 - ☐ Loop diuretics
 - ☐ Potassium-sparing diuretics
 - ☐ Beta-blockers
 - ☐ Direct renin inhibitors
 - ☐ Alpha-1 blockers
 - ☐ Central alpha-2 blockers
 - ☐ Direct vasodilators

Complications of Hypertension



CARDIOMYOPATHIES

- Cardiac diseases due to intrinsic myocardial dysfunction (literally, “heart muscle diseases”)
- Heart muscle disease as a result of intrinsic myocardial dysfunction
- These can be primary—that is, principally confined to the myocardium—or secondary presenting as the cardiac manifestation of a systemic disorder
- In many cases, the cardiomyopathy is of unknown etiology and thus is termed idiopathic; however, a number of previously “idiopathic” cardiomyopathies have been shown to be the consequence of specific genetic abnormalities

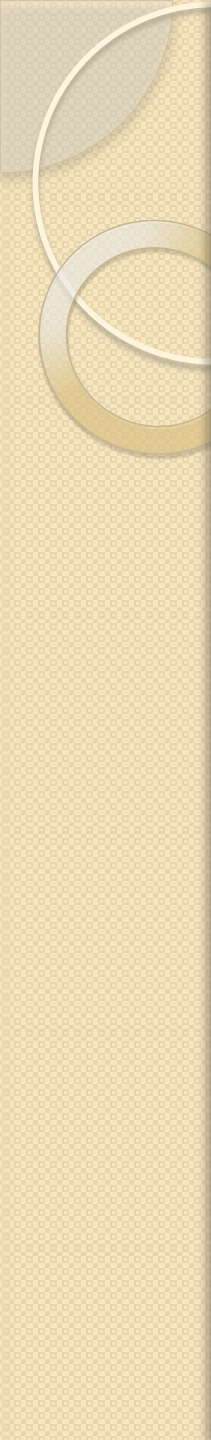
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- A more clinical and functional classification divides cardiomyopathies into three groups
 - ❑ Dilated cardiomyopathy (DCM) (including arrhythmogenic right ventricular cardiomyopathy)
 - ❑ Hypertrophic cardiomyopathy (HCM)
 - ❑ Restrictive cardiomyopathy
 - DCM is most common (90% of cases), and restrictive cardiomyopathy is the least frequent.

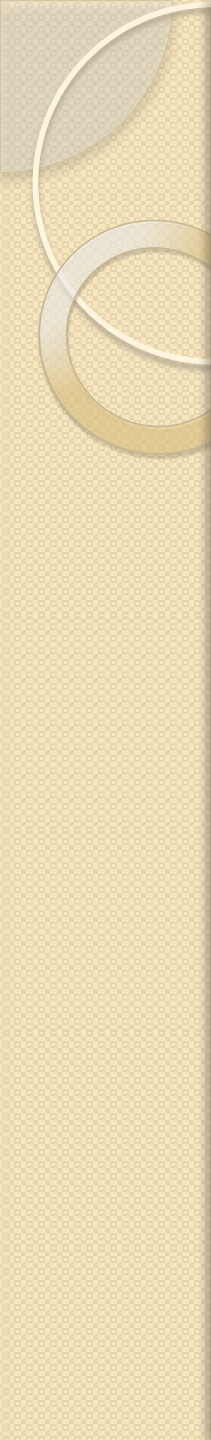
Functional Pattern	Left Ventricular Ejection Fraction*	Mechanisms of Heart Failure	Causes	Secondary Myocardial Dysfunction (Mimicking Cardiomyopathy)
Dilated	<40%	Impairment of contractility (systolic dysfunction)	Genetic; alcohol; peripartum; myocarditis; hemochromatosis; chronic anemia; doxorubicin (Adriamycin); sarcoidosis; idiopathic	Ischemic heart disease; valvular heart disease; hypertensive heart disease; congenital heart disease
Hypertrophic	50%–80%	Impairment of compliance (diastolic dysfunction)	Genetic; Friedreich ataxia; storage diseases; infants of diabetic mothers	Hypertensive heart disease; aortic stenosis
Restrictive	45%–90%	Impairment of compliance (diastolic dysfunction)	Amyloidosis; radiation-induced fibrosis; idiopathic	Pericardial constriction

*Range of normal values is approximately 50% to 65%.

Dilated Cardiomyopathy

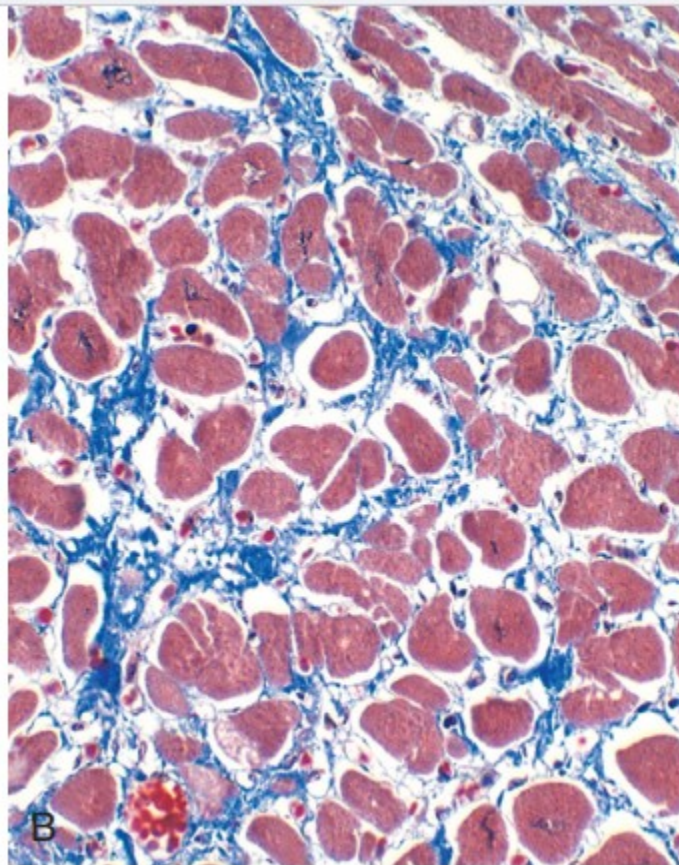
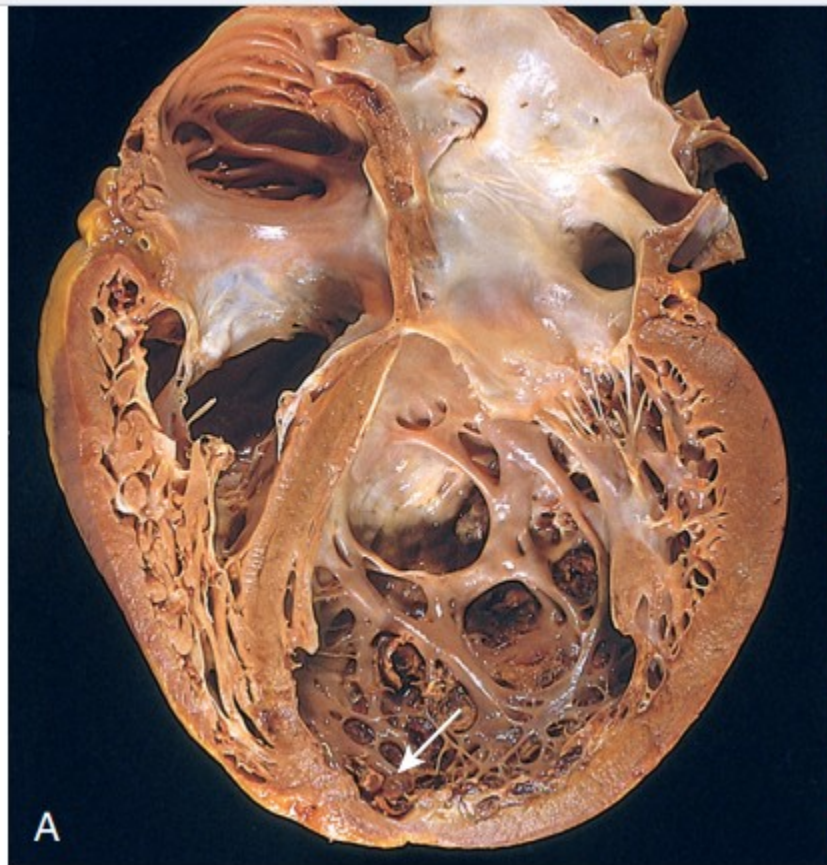
- Dilated cardiomyopathy (DCM) is characterized by progressive cardiac dilation and contractile (systolic) dysfunction, usually with concurrent hypertrophy
- Although many individuals with DCM have a familial (genetic) form, DCM can also result from various acquired myocardial insults.
- These include the following:
 - ❑ Myocarditis (an inflammatory disorder that precedes the development of cardiomyopathy in at least some cases, and is sometimes caused by viral infections)
 - ❑ Toxicities, including adverse effects of chemotherapeutic agents and chronic alcoholism, a history of which can be elicited in 10% to 20% of patients
 - ❑ Pregnancy, so-called “peripartum cardiomyopathy”
 - ❑ Stress-provoked
 - ❑ Tachycardia-induced

- 
- By the time it is diagnosed, DCM has frequently already progressed to end-stage disease; the heart is dilated and poorly contractile, and at autopsy or cardiac transplant, fails to reveal any specific pathologic features.

- 
- The heart in DCM characteristically is enlarged (up to two to three times the normal weight) and flabby, with dilation of all chambers
 - Because of the wall thinning that accompanies dilation, the ventricular thickness may be less than, equal to, or greater than normal.
 - Mural thrombi are often present and may be a source of thromboemboli.
 - By definition, valvular and vascular lesions (e.g., atherosclerotic coronary artery disease) that can cause cardiac dilation secondarily are absent.
 - The characteristic histologic abnormalities in DCM are nonspecific.

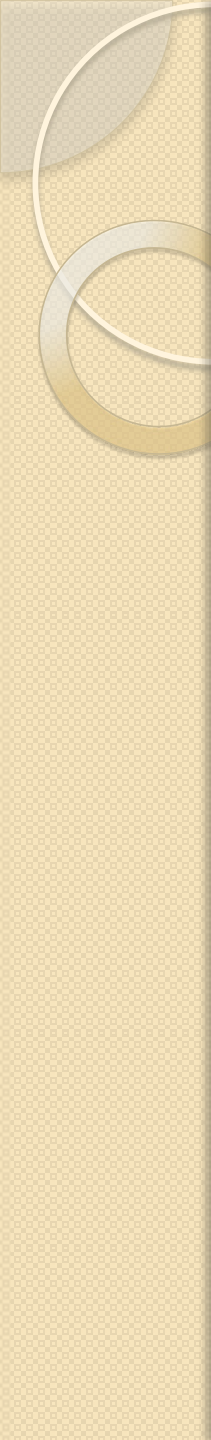
Clinical Features

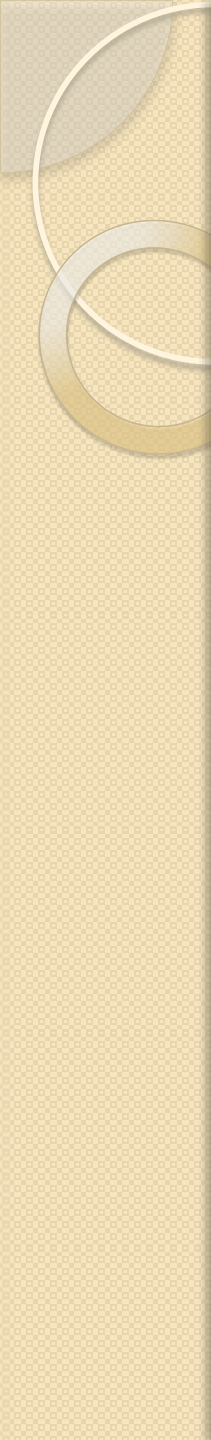
- The fundamental defect in DCM is ineffective contraction.
- Thus, in end-stage DCM, the cardiac ejection fraction typically is less than 25% (normal is 50% to 65%).
- Secondary mitral regurgitation and abnormal cardiac rhythms
- DCM most commonly is diagnosed between 20 and 50 years of age.
- It typically manifests with signs of slowly progressive CHF, including dyspnea, easy fatigability, and poor exertional capacity.
- One half of patients die within 2 years and only 25% survive longer than 5 years; death is usually due to progressive cardiac failure or arrhythmia.

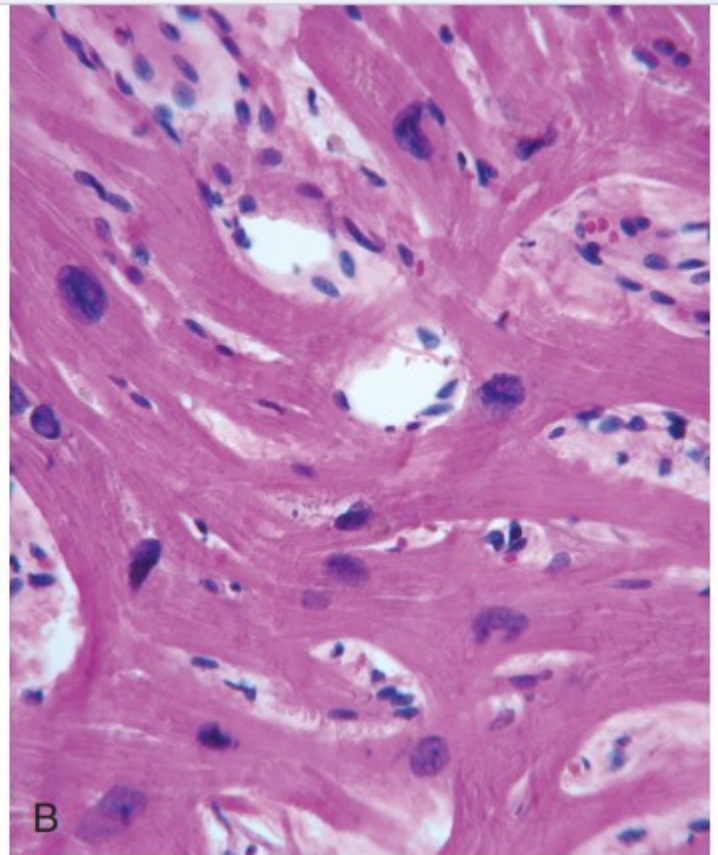
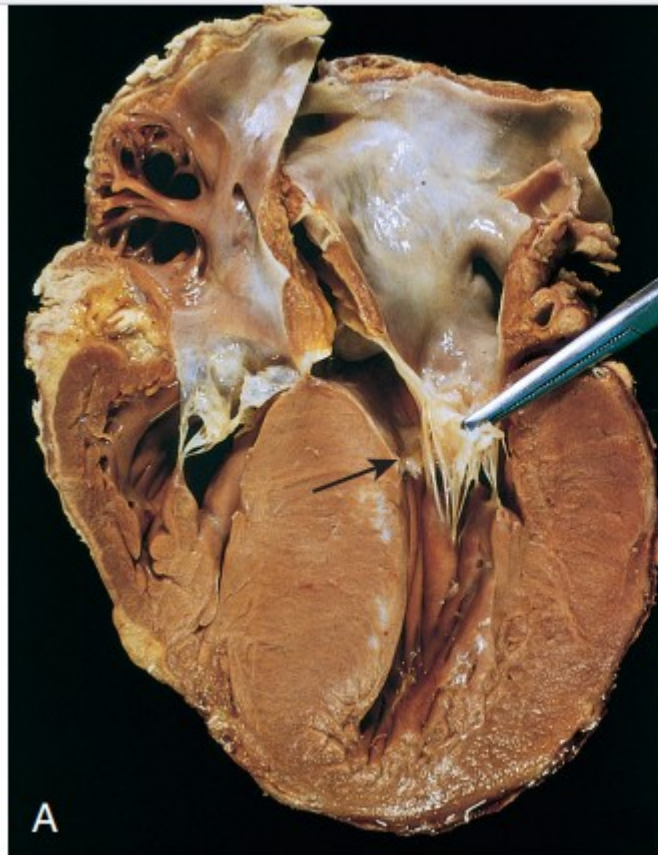


Hypertrophic Cardiomyopathy

- Is characterized by myocardial hypertrophy, abnormal diastolic filling, and-in a third of cases-ventricular outflow obstruction
- The heart is thick-walled, heavy, and hyper contractile
- Point mutation in genes encoding sarcomeric proteins
- Massive myocardial hypertrophy without ventricular dilation
- Classically, there is disproportionate thickening of the ventricular septum relative to the left ventricle free wall (so-called asymmetric septal hypertrophy)
- Impaired diastolic filling (small chamber size), Reduced stroke volume
- Exertional dyspnea ,myocardial ischemia

- 
- The characteristic histologic features in HCM are marked myocyte hypertrophy, haphazard myocyte (and myofiber) disarray, and interstitial fibrosis
 - Major clinical problems include atrial fibrillation with mural thrombus formation, IE of the mitral valve, CHF, arrhythmias, and sudden death

- 
- Most patients' symptoms are improved by therapy that promotes ventricular relaxation; partial surgical excision or controlled alcohol-induced infarction of septal muscle also can relieve the outflow tract obstruction.
 - As mentioned earlier, HCM is an important cause of sudden cardiac death.
 - In almost one third of cases of sudden cardiac death in athletes younger than 35 years of age, the underlying cause is HCM



Restrictive Cardiomyopathy

- Characterized by a primary decrease in ventricular compliance, resulting in impaired ventricular filling during diastole (simply put, the wall is stiffer)
- Least common
- In idiopathic restrictive cardiomyopathy the ventricles are of approximately normal size or slightly enlarged, the cavities are not dilated, and the myocardium is firm.
- Radiation fibrosis, amyloidosis, sarcoidosis, or products of inborn errors of metabolism.
- Biatrial dilation is commonly observed.
Microscopically there is interstitial fibrosis, varying from minimal and patchy to extensive and diffuse